

LA's BeST @USC

Spatial Transcriptomics

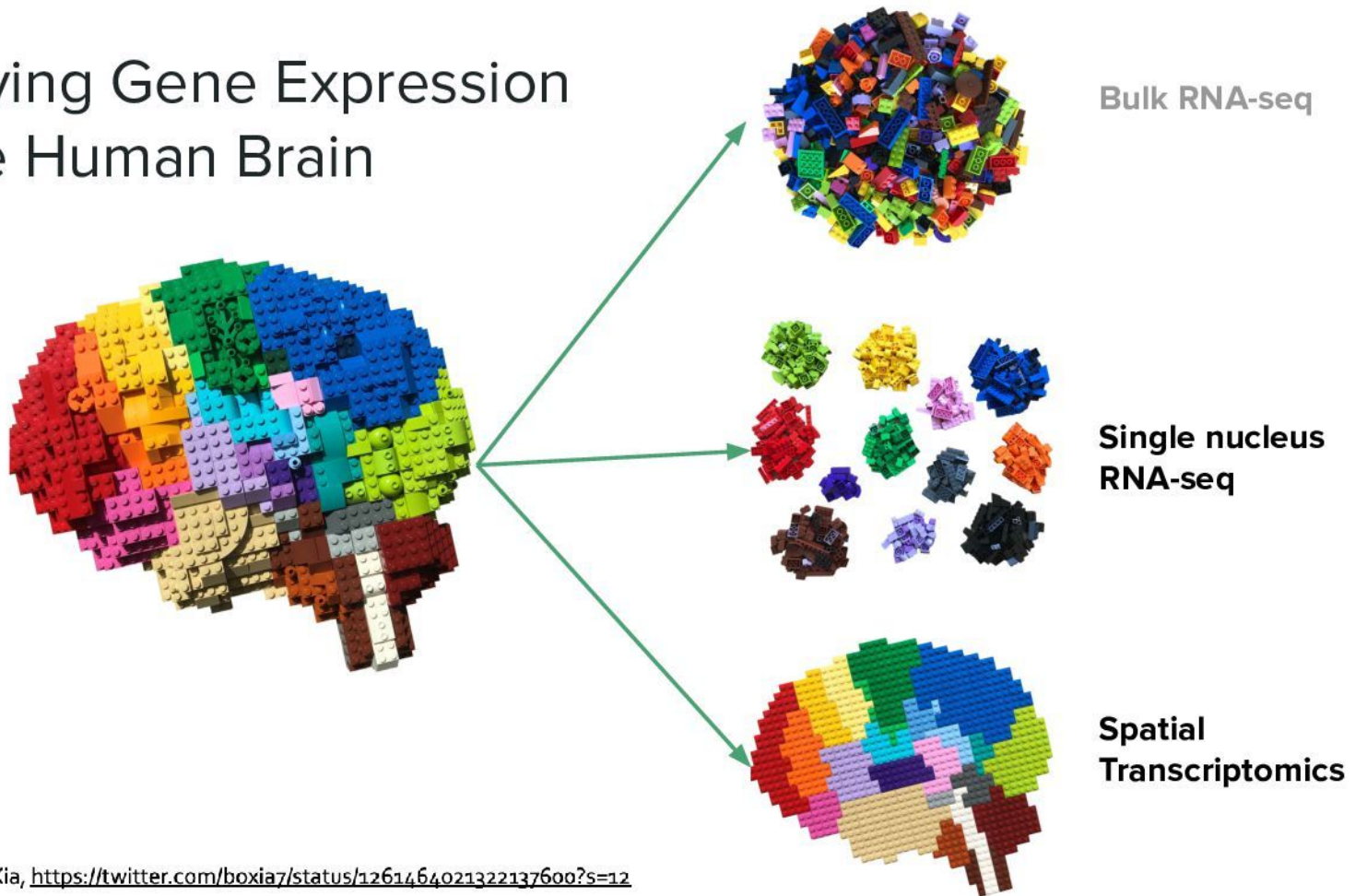
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Southern California

Studying Gene Expression in the Human Brain



REVIEW ARTICLE

<https://doi.org/10.1038/s41592-022-01409-2>

nature | methods



Museum of spatial transcriptomics

Lambda Moses¹ and Lior Pachter^{1,2}  

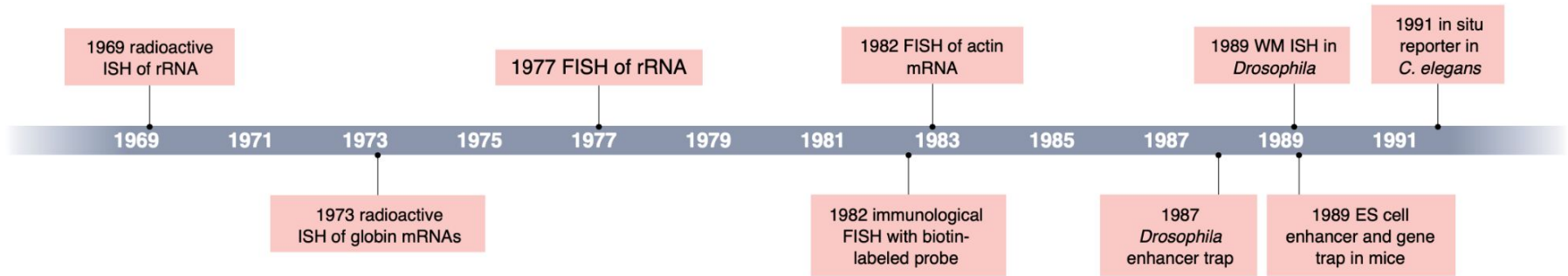
The function of many biological systems, such as embryos, liver lobules, intestinal villi, and tumors, depends on the spatial organization of their cells. In the past decade, high-throughput technologies have been developed to quantify gene expression in space, and computational methods have been developed that leverage spatial gene expression data to identify genes with spatial patterns and to delineate neighborhoods within tissues. To comprehensively document spatial gene expression technologies and data-analysis methods, we present a curated review of literature on spatial transcriptomics dating back to 1987, along with a thorough analysis of trends in the field, such as usage of experimental techniques, species, tissues studied, and computational approaches used. Our Review places current methods in a historical context, and we derive insights about the field that can guide current research strategies. A companion supplement offers a more detailed look at the technologies and methods analyzed: https://pachterlab.github.io/LP_2021/.

It has long been recognized that in biological systems ranging from the *Drosophila* embryo to the hepatic lobule, many genes need to be properly regulated in space for the system to function. To study the spatial patterns of gene expression, many different spatial transcriptomics methods, which produce spatially localized quantification of messenger RNA (mRNA) transcripts as proxies for gene expression, have been developed. Thanks to growing interest in the field, several reviews have been written in the past 5 years, providing overviews

Prequel era

By “spatial transcriptomics”, we mean attempts to quantify mRNA expression of large numbers of genes within the spatial context of tissues and cells. Some important technologies enabling spatial transcriptomics date back to the 1970s (Chapter 2 in Supplementary Information). Various forms of in situ hybridization (ISH) have been used for a long time to visualize gene expression in space. Radioactive ISH was first introduced in 1969, visualizing ribosomal

Spatial Biology Technologies



Prequel



ROI selection



NGS barcoding



smFISH



ISS

Spatial Biology Technologies

In situ hybridization

🌐 14 languages ▾

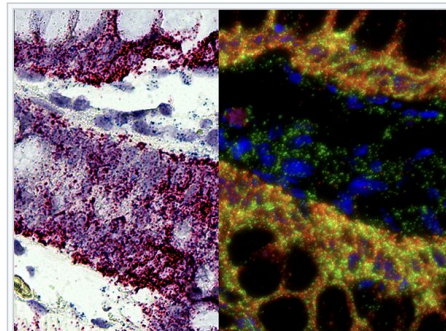
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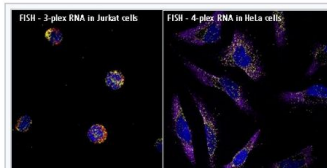
From Wikipedia, the free encyclopedia

***In situ* hybridization (ISH)** is a type of [hybridization](#) that uses a labeled [complementary DNA](#), [RNA](#) or modified nucleic acid strand (i.e., a [probe](#)) to localize a specific DNA or RNA sequence in a portion or section of [tissue](#) (*in situ*) or if the tissue is small enough (e.g., plant seeds, *Drosophila* embryos), in the entire tissue (whole mount ISH), in cells, and in [circulating tumor cells](#) (CTCs). This is distinct from [immunohistochemistry](#), which usually localizes proteins in tissue sections.

In situ hybridization is used to reveal the location of specific nucleic acid sequences on chromosomes or in tissues, a crucial step for understanding the organization, regulation, and function of genes. The key techniques currently in use include *in situ* hybridization to mRNA with [oligonucleotide](#) and RNA probes (both radio-labeled and hapten-labeled), analysis with light and electron microscopes, whole mount *in situ* hybridization, double detection of RNAs and RNA plus protein, and fluorescent *in situ* hybridization to detect chromosomal sequences. DNA ISH can be used to determine the [structure](#) of chromosomes. [Fluorescent DNA ISH \(FISH\)](#) can, for example, be used in medical diagnostics to assess chromosomal integrity. RNA ISH (RNA *in situ* hybridization) is used to measure and localize RNAs (mRNAs, lncRNAs, and miRNAs) within tissue sections, cells, whole mounts, and circulating tumor cells (CTCs). *In situ* hybridization was invented by American biologists [Mary-Lou Pardue](#) and [Joseph G. Gall](#).^{[1][2][3]}



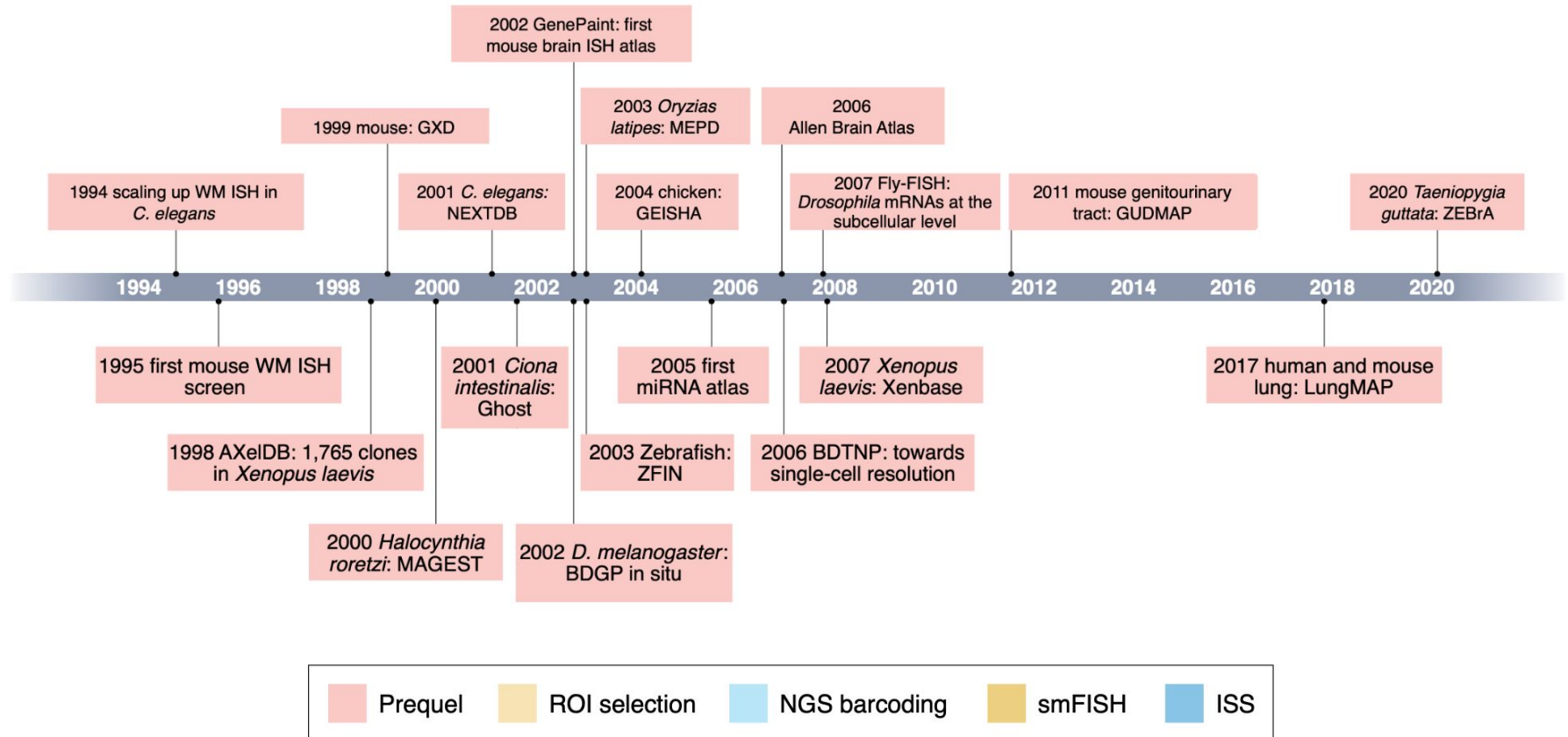
RNA *in situ* hybridization - [KRT5](#) and housekeeping gene in human [melanoma FFPE](#) tissue section - visualized under brightfield and fluorescence microscope



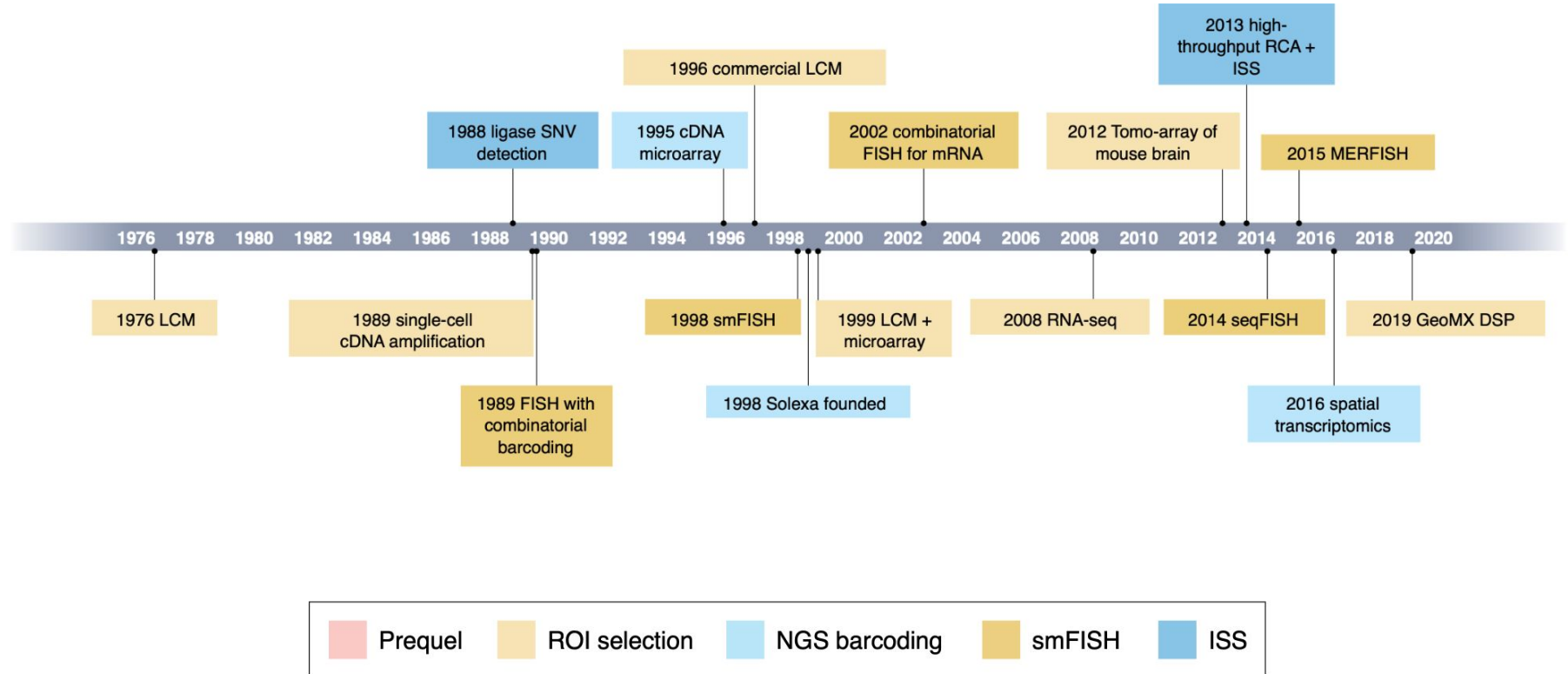
Multiplex RNA visualization in cells using ViewRNA FISH Assays

Challenges of in-situ hybridization [\[edit\]](#)

Spatial Biology Technologies



Spatial Biology Technologies



Spatial Biology Technologies

Method	First published	Category	Max # genes	Min spot diameter (μ m)
voxelation	2002-01-31	ROI selection	Tx wide	NA
PA-GFP	2010-11-11	ROI selection	Tx wide	NA
SRM seqFISH	2012-06-02	smFISH	32	single cell
Tomo-array	2012-09-18	ROI selection	Tx wide	NA
iceFISH	2013-02-16	smFISH	20	single cell
ISS	2013-07-14	ISS	256	single cell
...	...	...	...	...

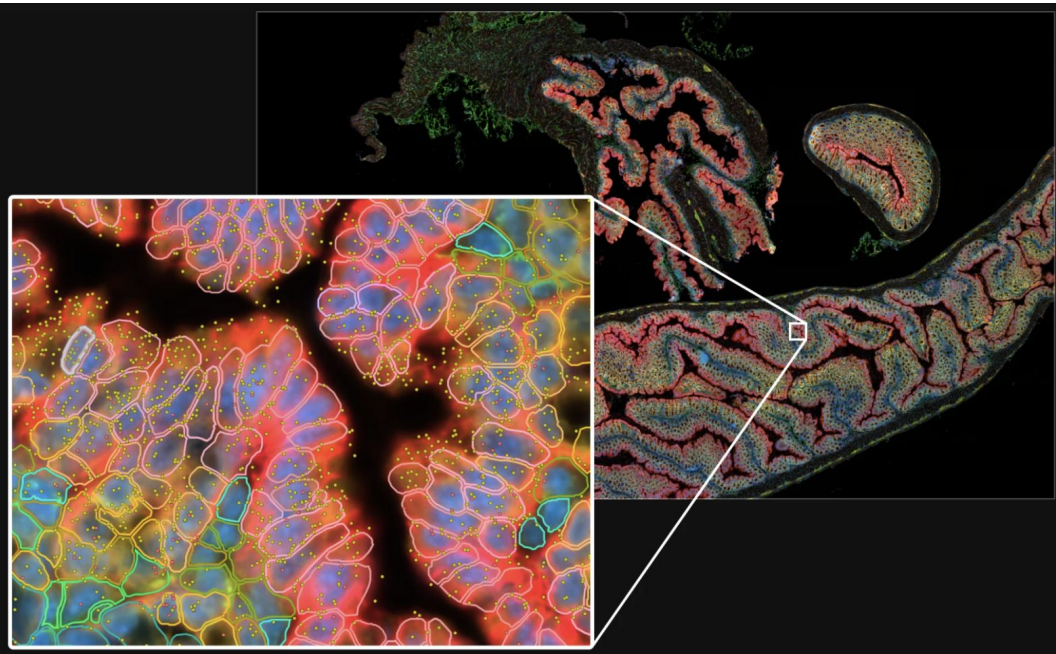
203 entries!

Spatial Biology Technologies

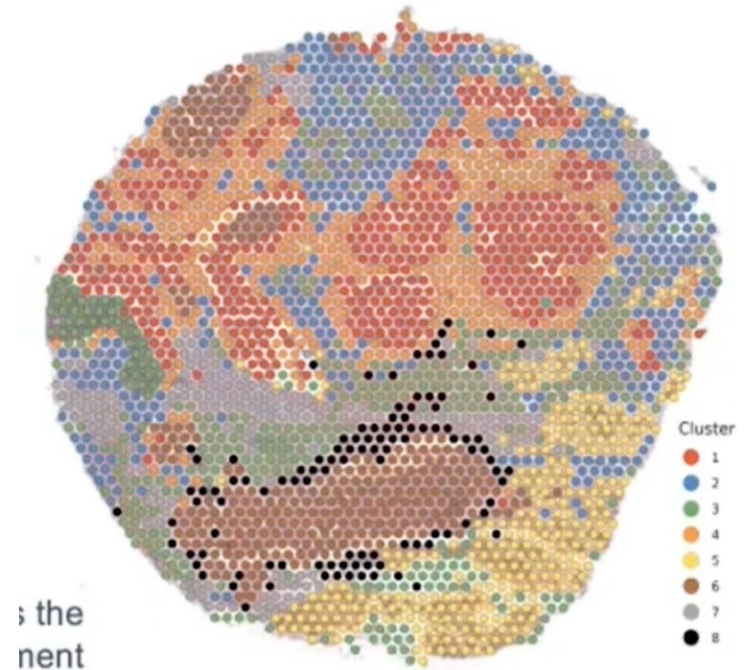
- seqFISH
- MERFISH
- IMC
- CosMx
- 10X Genomics:
 - Visium
 - Xenium
- many others...

Spatial Biology Technologies

Imaging



Sequencing



Spatial Biology Technologies

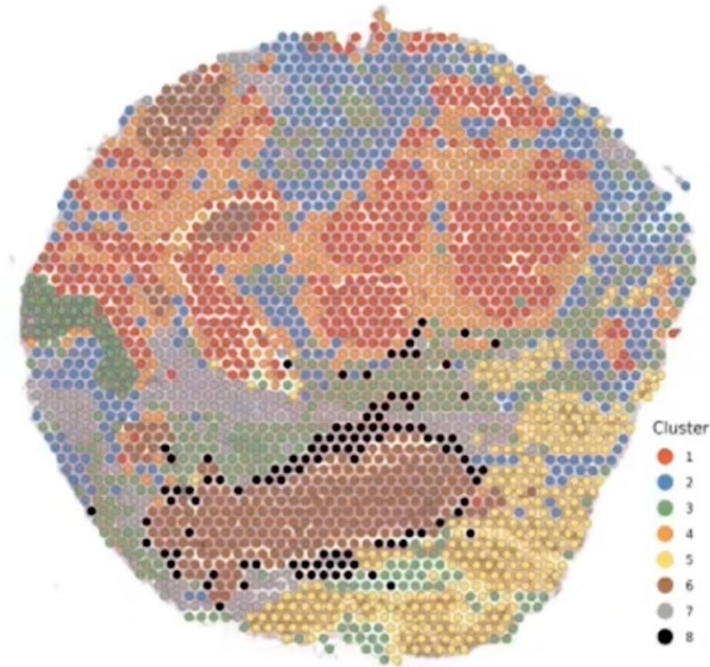
Features:

- Transcriptome
- Select RNAs
- Select proteins

Observations:

- Spots
- Cells
- Molecules

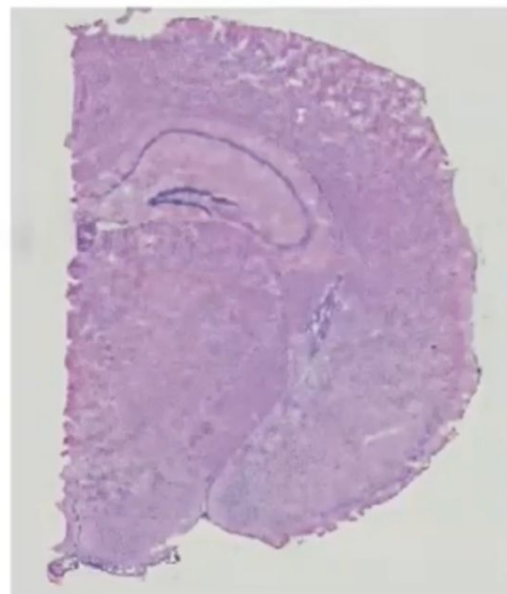
Sequencing-Based Spatial Transcriptomics



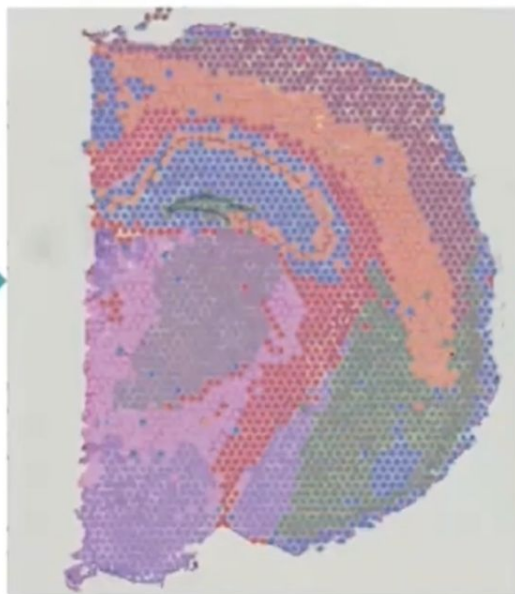
Two types of information comprise Visium data

Tissue image and gene expression

Histology image



Histology + Gene expression



Gene expression

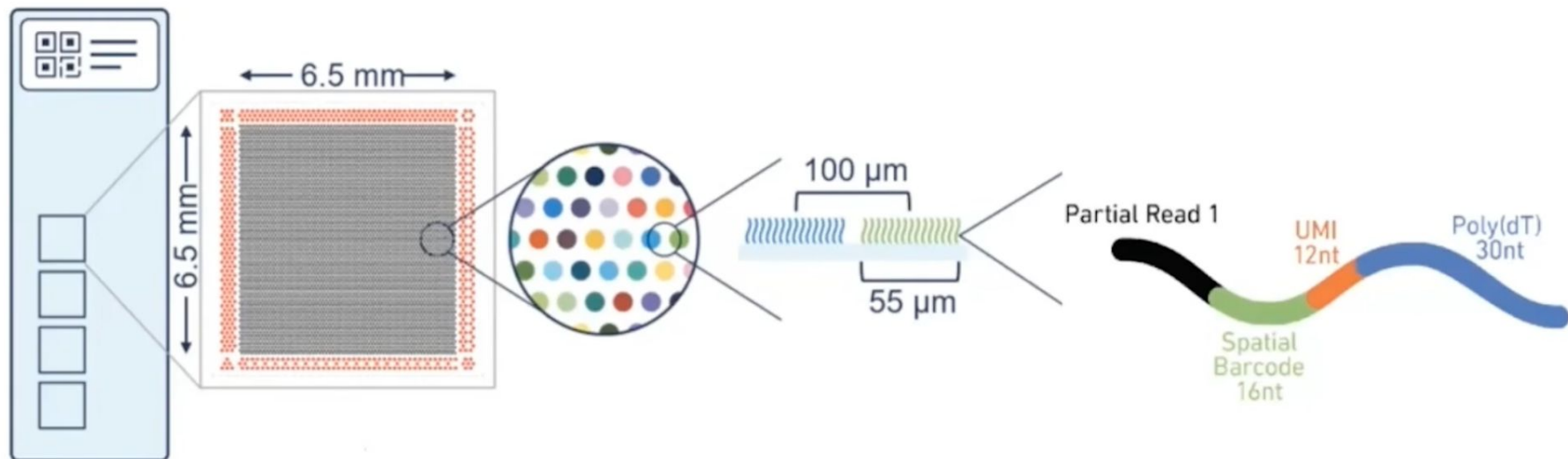


Spatial capture technology for whole transcriptome analysis

Visium Spatial Gene Expression Slide

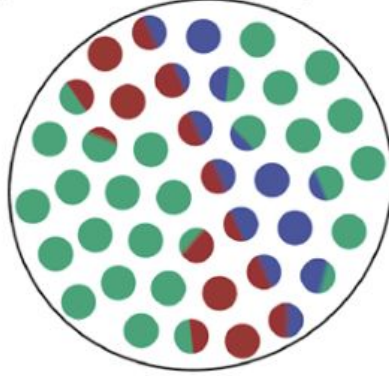
Capture Area with ~5000 Barcoded Spots

Visium Gene Expression Barcoded Spots

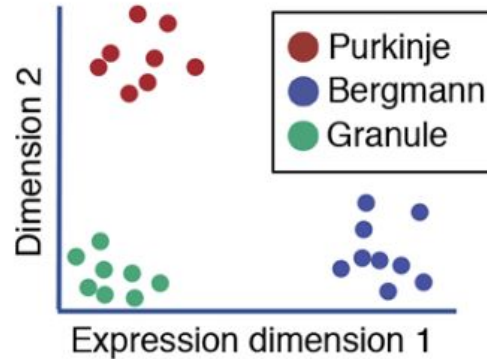


Spots are not cells

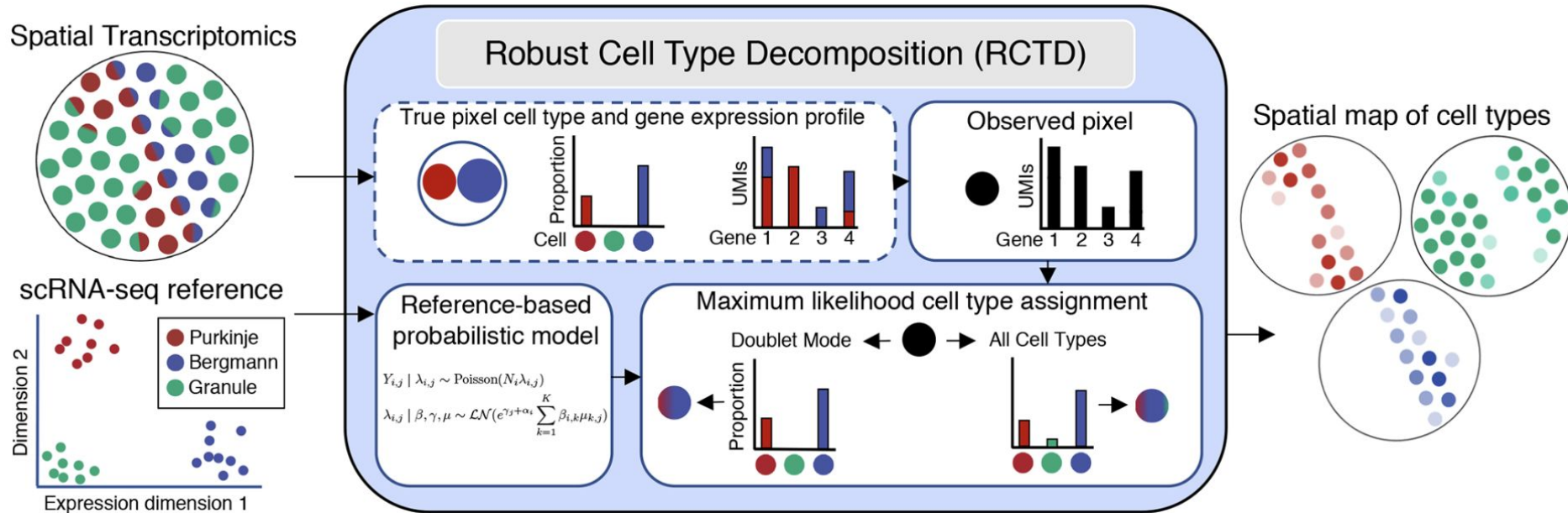
Spatial Transcriptomics



scRNA-seq reference



Deconvolution



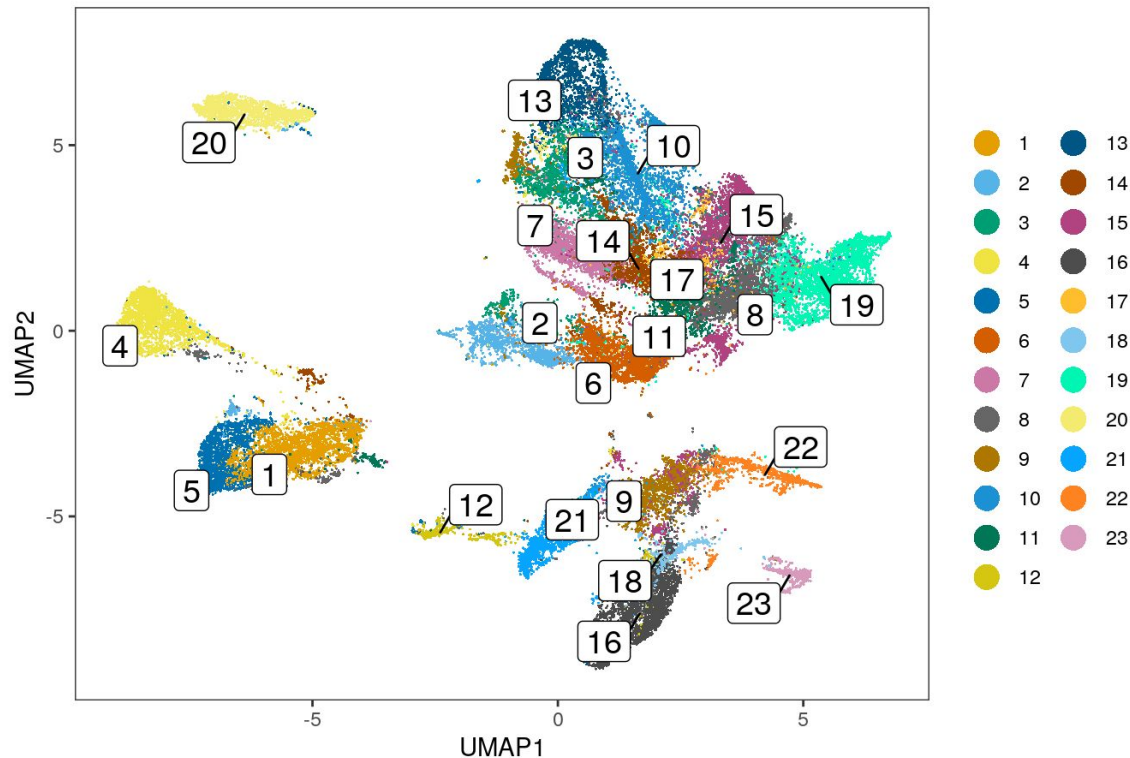
“Classic” Single-Cell Analysis

Highly Variable Genes

Dimensionality Reduction

Clustering

Differential Expression



Spatial Transcriptomics Analysis

Spatially Informed Highly Variable Genes

Spatially Informed Dimensionality Reduction

Spatially Informed Clustering

Spatially Informed Differential Expression

Spatial Transcriptomics Analysis

How much do we actually want this?

Spatially Informed Highly Variable Genes

Spatially Informed Dimensionality Reduction

Spatially Informed Clustering

Spatially Informed Differential Expression

Highly Variable Genes

```
FindVariableFeatures()
```

“We next calculate a subset of features that exhibit **high cell-to-cell variation** in the dataset (i.e, they are highly expressed in some cells, and lowly expressed in others). We and others have found that focusing on these genes in downstream analysis helps to **highlight biological signal** in single-cell datasets.”

Highly Variable Genes

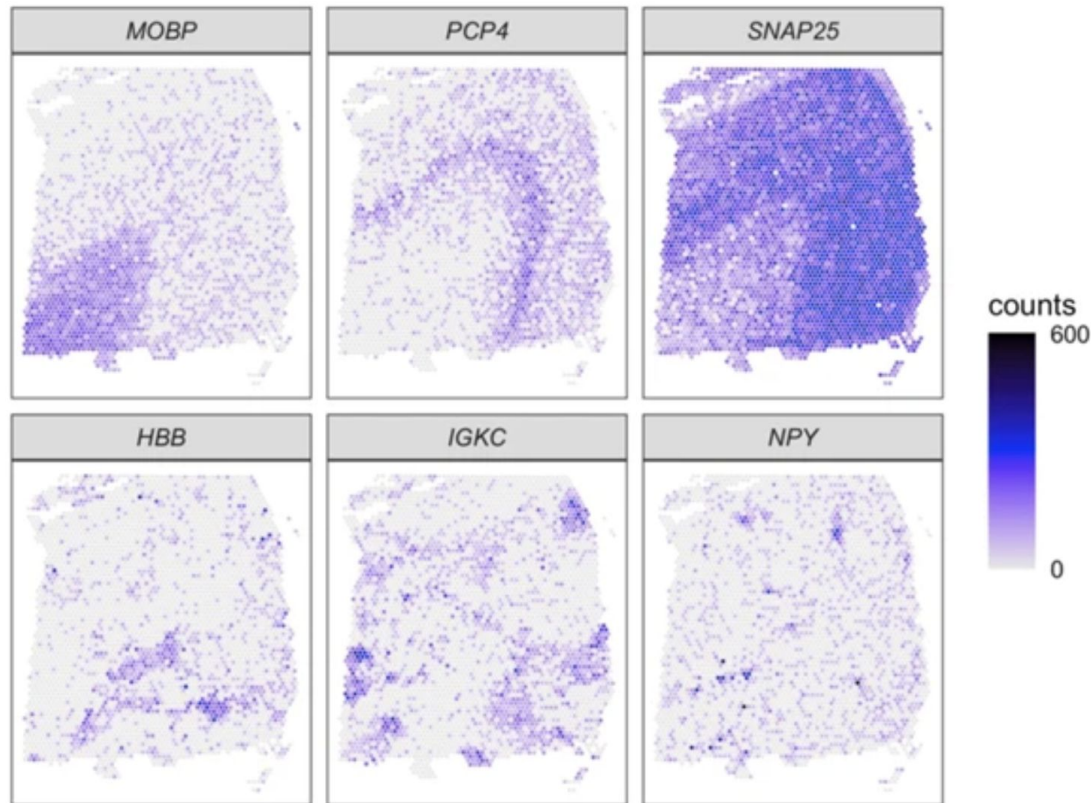
```
FindVariableFeatures()
```

“We next calculate a subset of features that exhibit **high cell-to-cell variation** in the dataset (i.e, they are highly expressed in some cells, and lowly expressed in others). We and others have found that focusing on these genes in downstream analysis helps to **highlight biological signal** in single-cell datasets.”

It also just makes the computation a lot easier
Cuts down on multiple testing

Spatially Variable Genes

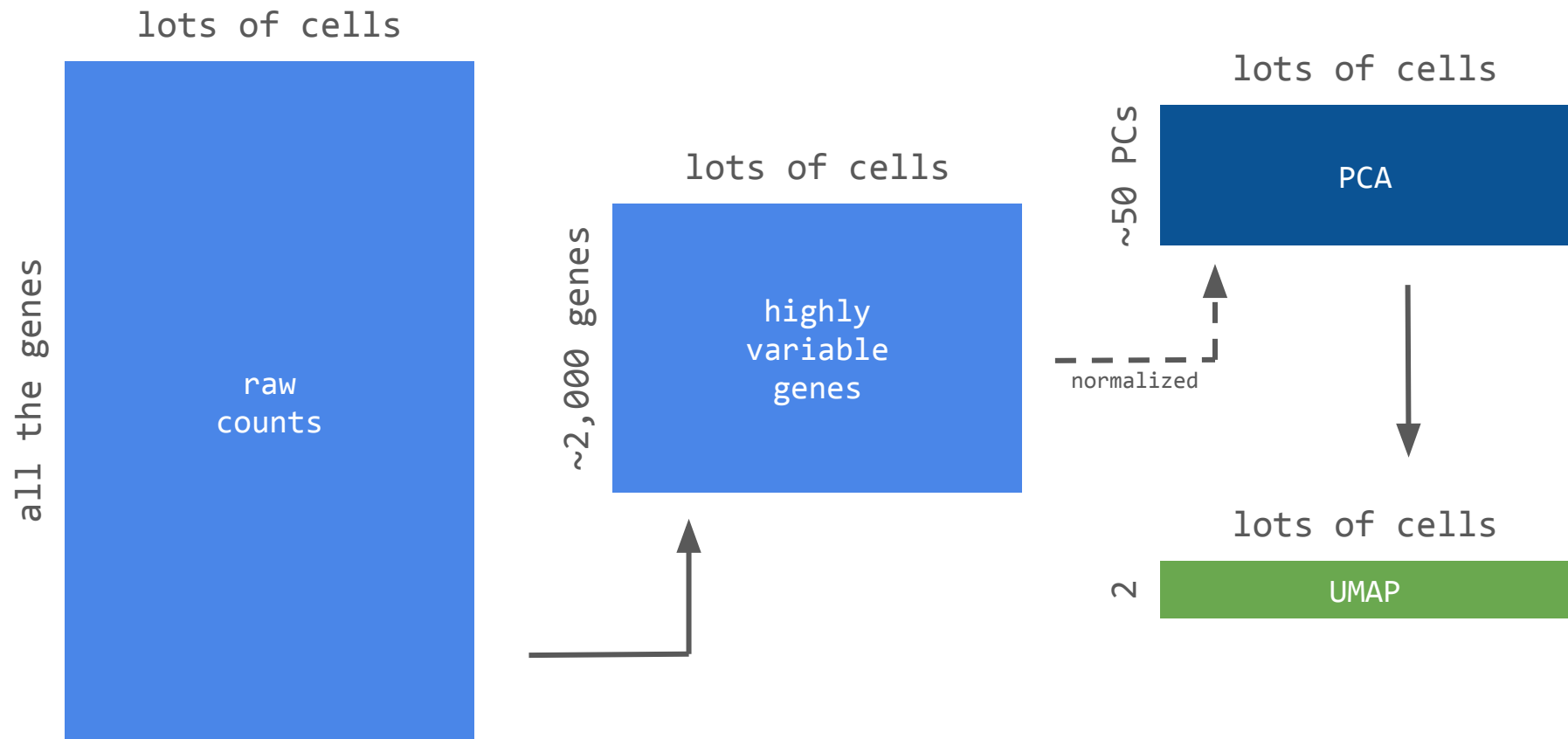
Selected SVGs: human DLPFC



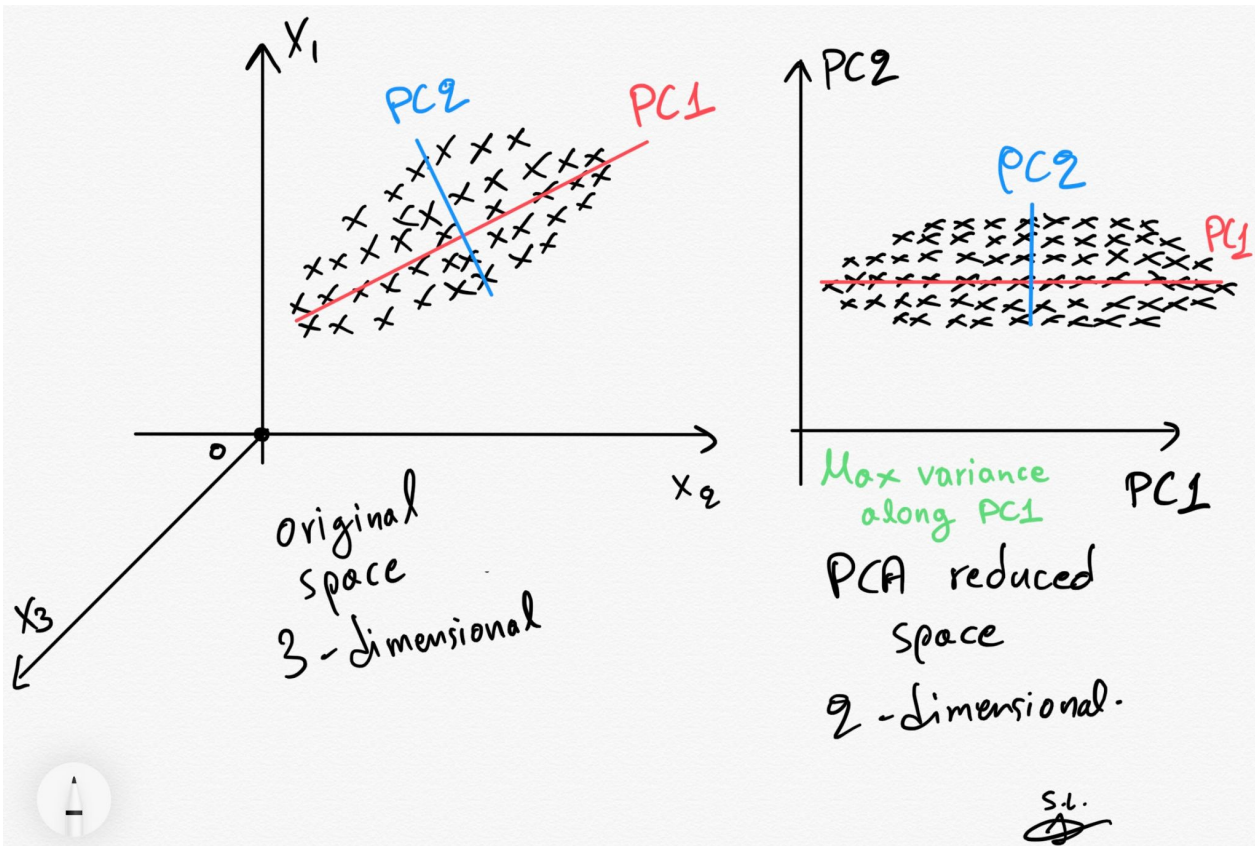
{nnSVG}

- Rank genes by (spatial) variability
 - Select the top N genes
- + takes advantage of spatial information
- could struggle with very disperse cell types

Dimensionality Reduction



PCA



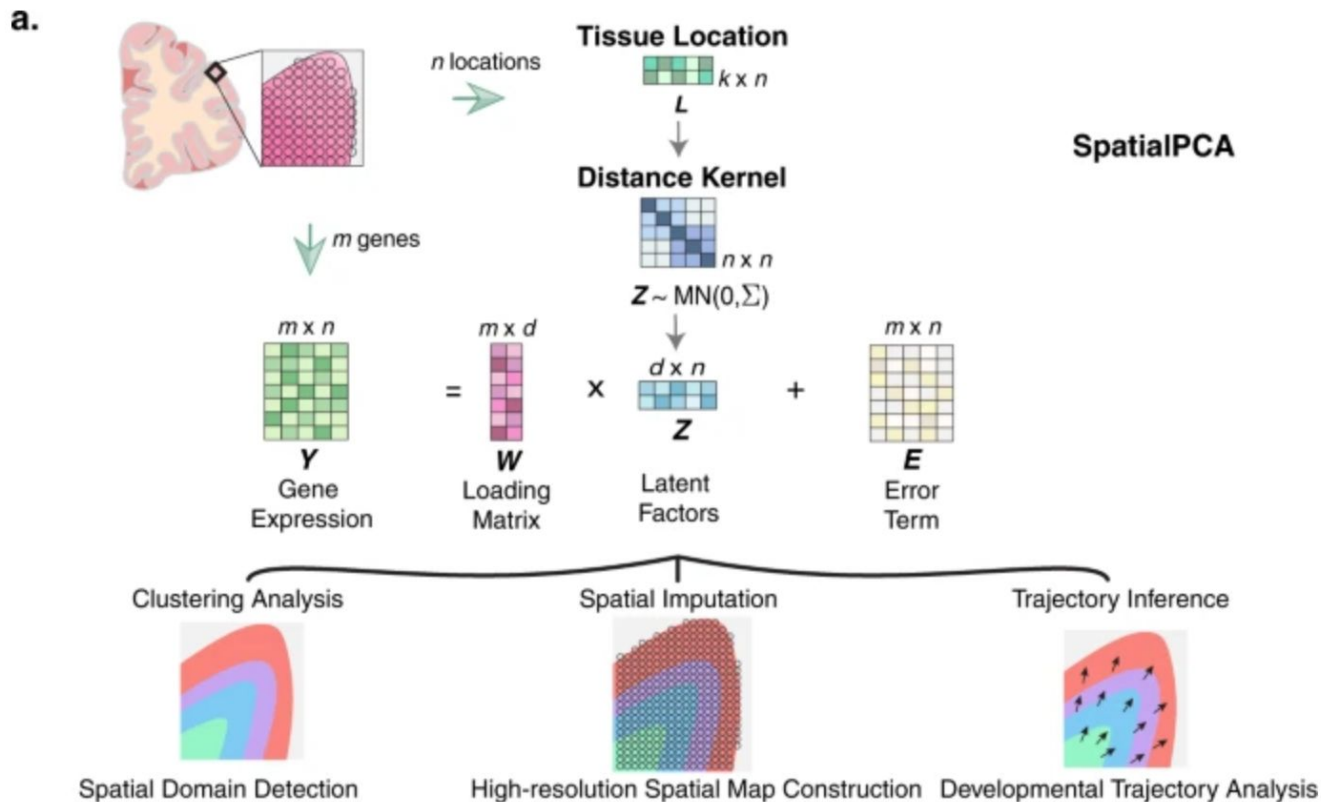
Distills the signal(s) in the data

Can identify multiple axes of variation

Further reduces computational complexity

SpatialPCA

Fig. 1: Method schematic of SpatialPCA and simulation results.



Dimensionality Reduction

How much should dimensionality reduction be determined by spatial positioning vs. gene expression?

spatially driven



expression driven

just use spatial
coordinates

SpatialPCA

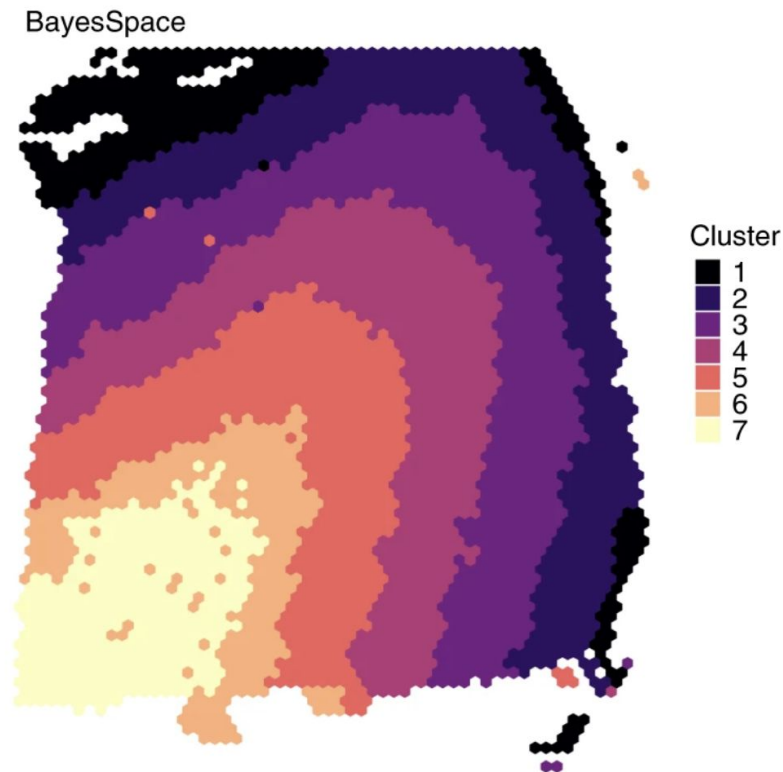
PCA (ignore spatial
coordinates)

Clustering / Domain Detection

Clustering finds **groups of cells** with similar gene expression patterns.

Domain detection finds **spatial regions** with similar gene expression patterns.

Do you want groups driven more by spatial organization or by gene expression?



Clustering / Domain Detection

How much should clustering be determined by spatial positioning vs. gene expression?

spatially driven



expression driven

draw partitions
on spatial
coordinates

BayesSpace

Louvain clustering
(ignore spatial
coordinates)

Clustering / Domain Detection

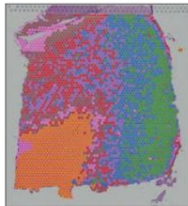
Ground truth
(Data9; slice 151673)



Domains

- Layer 1
- Layer 2
- Layer 3
- Layer 4
- Layer 5
- Layer 6
- White matter

Louvain
NMI 0.303



Leiden
NMI 0.307



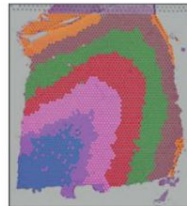
SpaGCN
NMI 0.491



SpaGCN(HE)
NMI 0.478



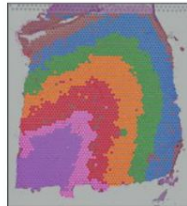
BayesSpace
NMI 0.688



stLearn
NMI 0.548



SEDR
NMI 0.631



CCST
NMI 0.449



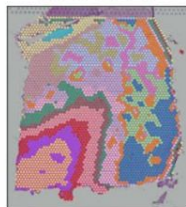
SCAN-IT
NMI 0.577



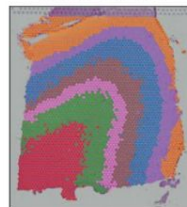
STAGATE
NMI 0.592



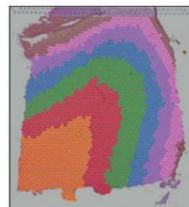
SpaceFlow
NMI 0.447



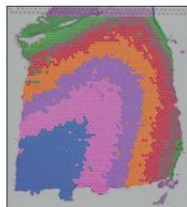
conST
NMI 0.672



BASS
NMI 0.71

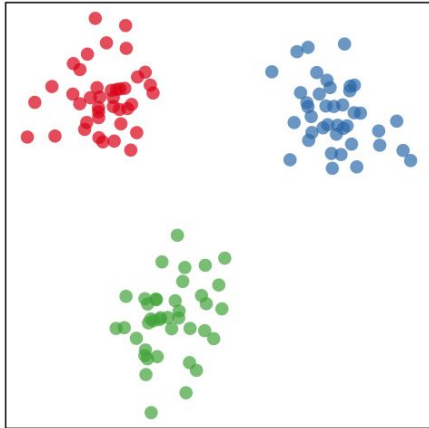


GraphST
NMI 0.697

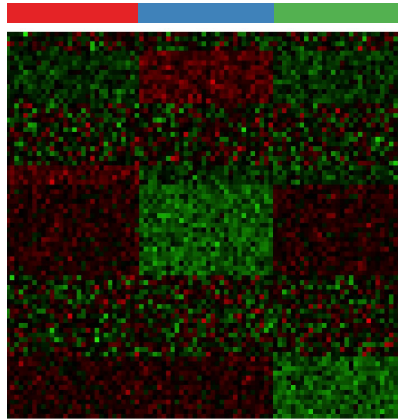


Downstream Analysis

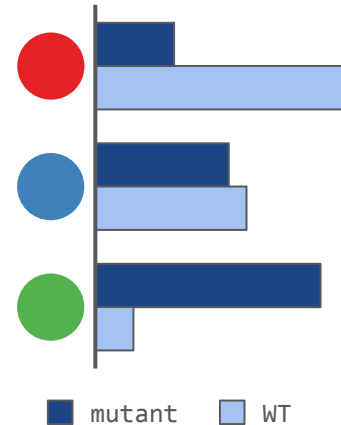
Clustering



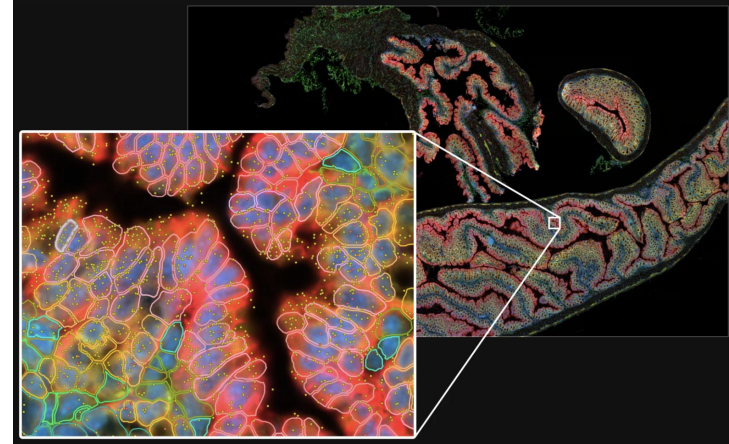
Differential Expression



Differential Abundance

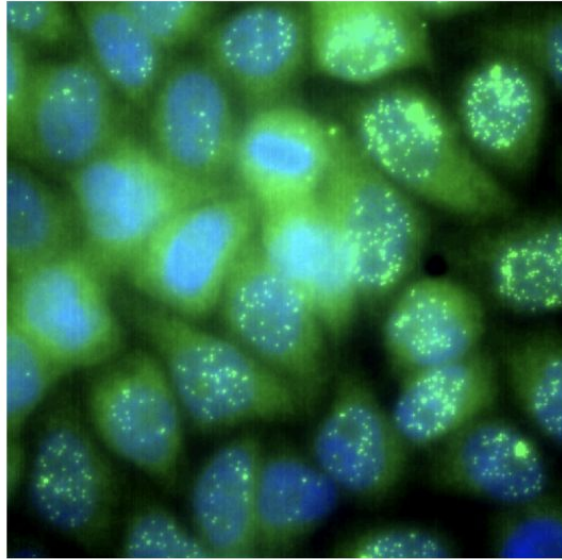


Imaging-Based Spatial Biology



Raw Data

original image



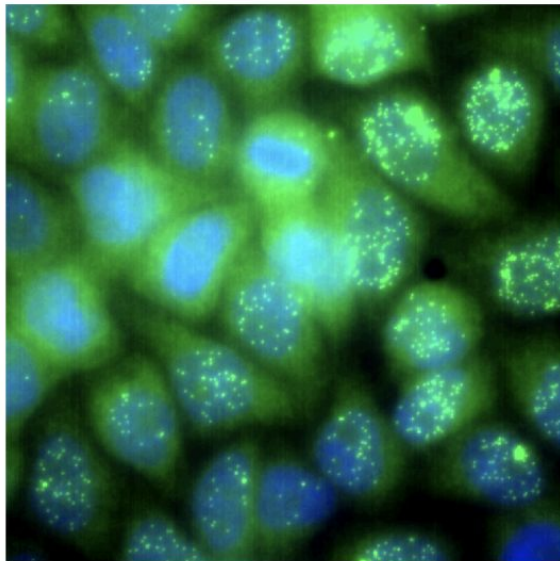
Collection of images

One channel (color)
per gene/protein

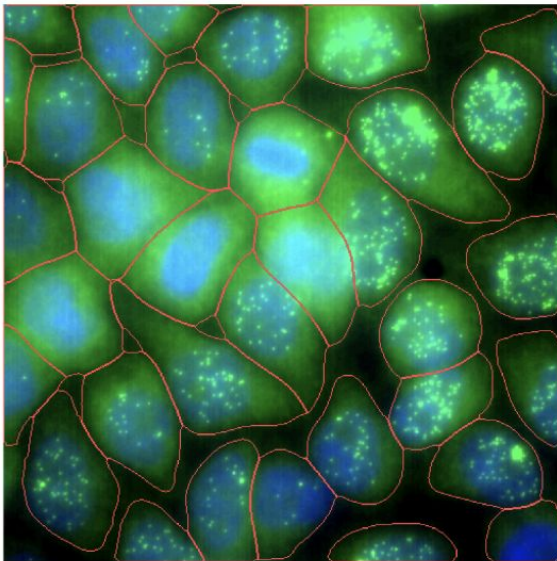
Often very large

Segmentation

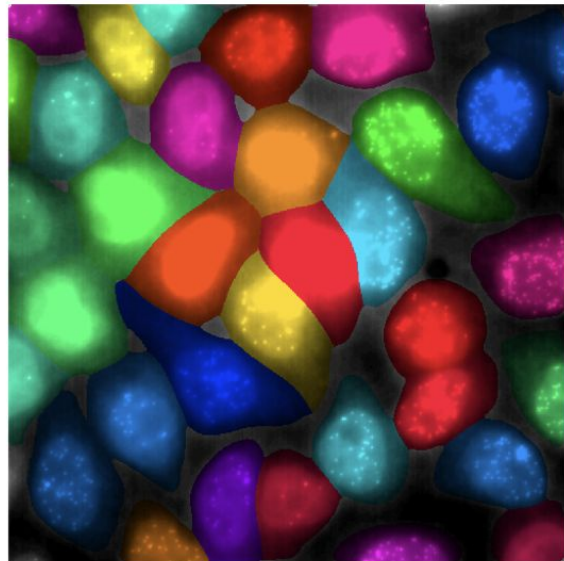
original image



predicted outlines

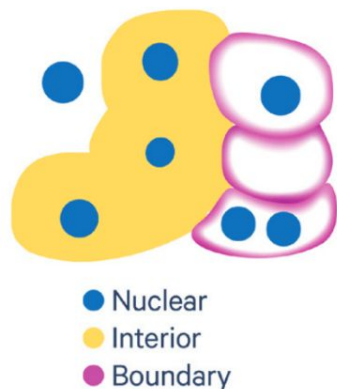


predicted masks



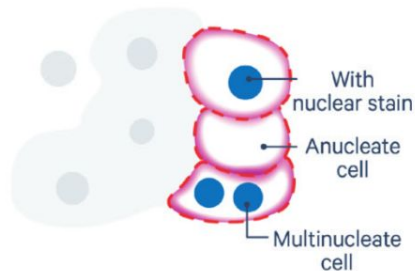
10x Space Ranger / Xenium

Types of Stains

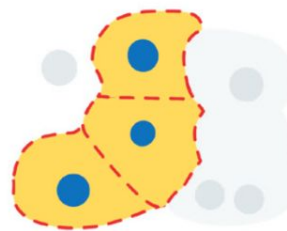


Types of Cell Segmentation

Boundary stain
(does not require nuclear staining)

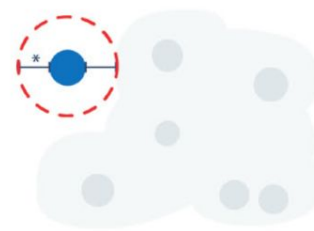


Interior stain and expansion
(requires nuclear staining)



----- Inferred cell segmentation

Nuclear expansion
(requires nuclear staining)



*Expands 5 μ m or until another cell boundary is encountered

Figure 3. A representation of the Xenium multimodal cell segmentation algorithm.

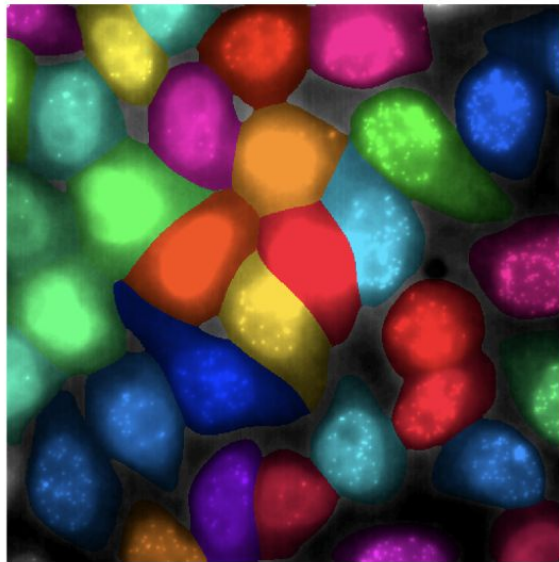
Segmentation

Can compress to one
image layer

Transcripts/proteins
assigned to cells
(observations)

Metadata on cell shape

predicted masks



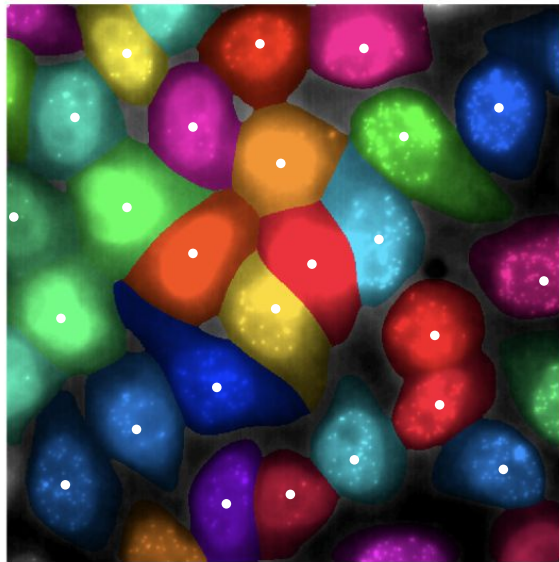
Segmentation

Can compress to one
image layer

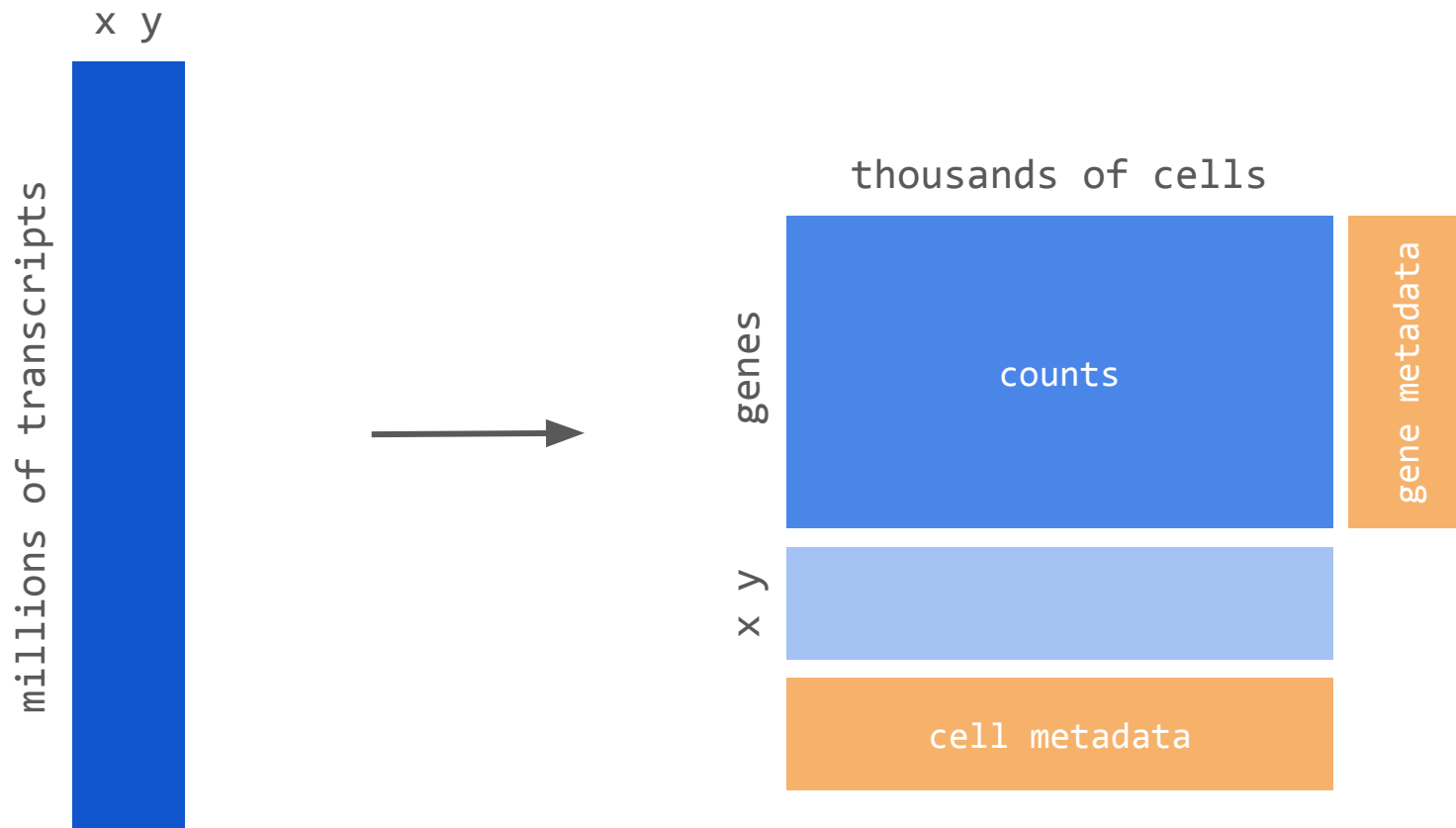
Transcripts/proteins
assigned to cells
(observations)

Metadata on cell shape

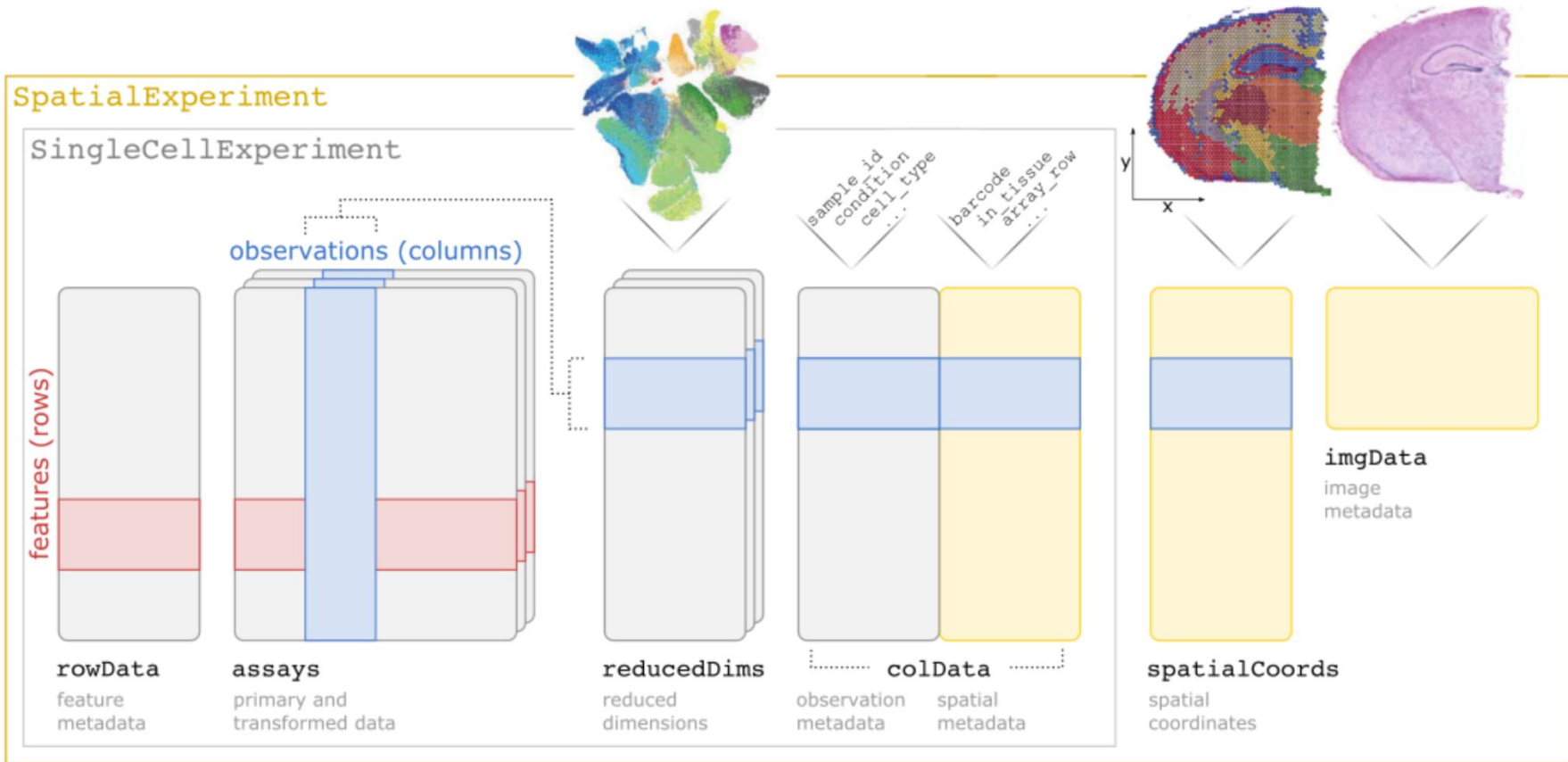
predicted masks



Data Structure



Data Structure - SpatialExperiment



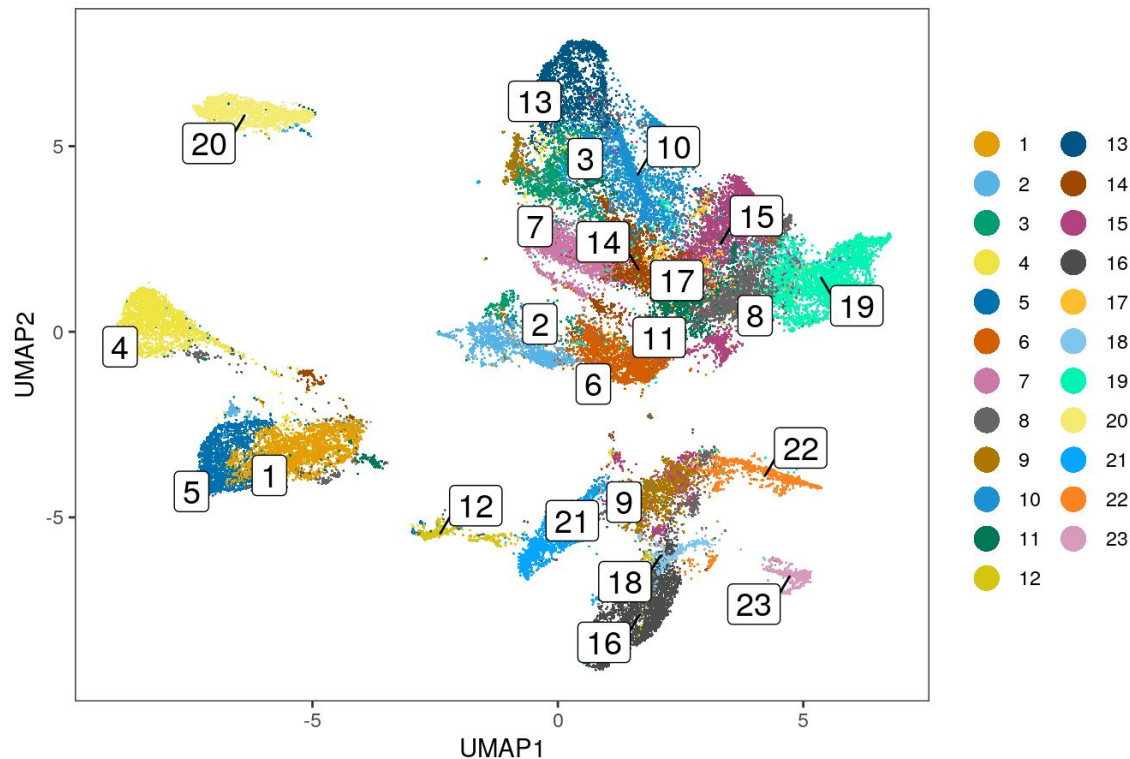
“Classic” Single-Cell Analysis

Highly Variable Genes

Dimensionality Reduction

Clustering

Differential Expression



Imaging-Based Spatial Analysis

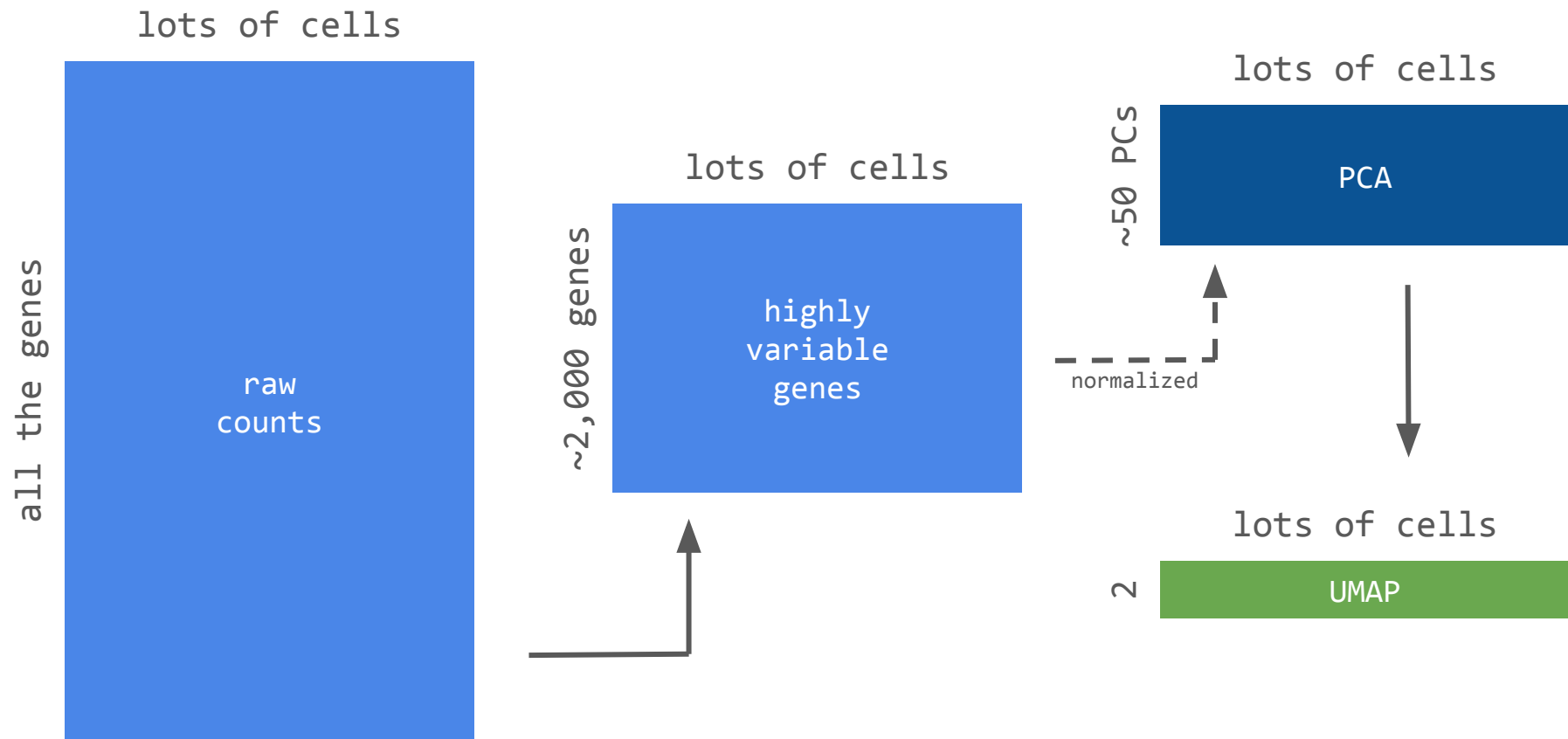
~~Highly Variable Genes~~

Spatially Informed Dimensionality Reduction

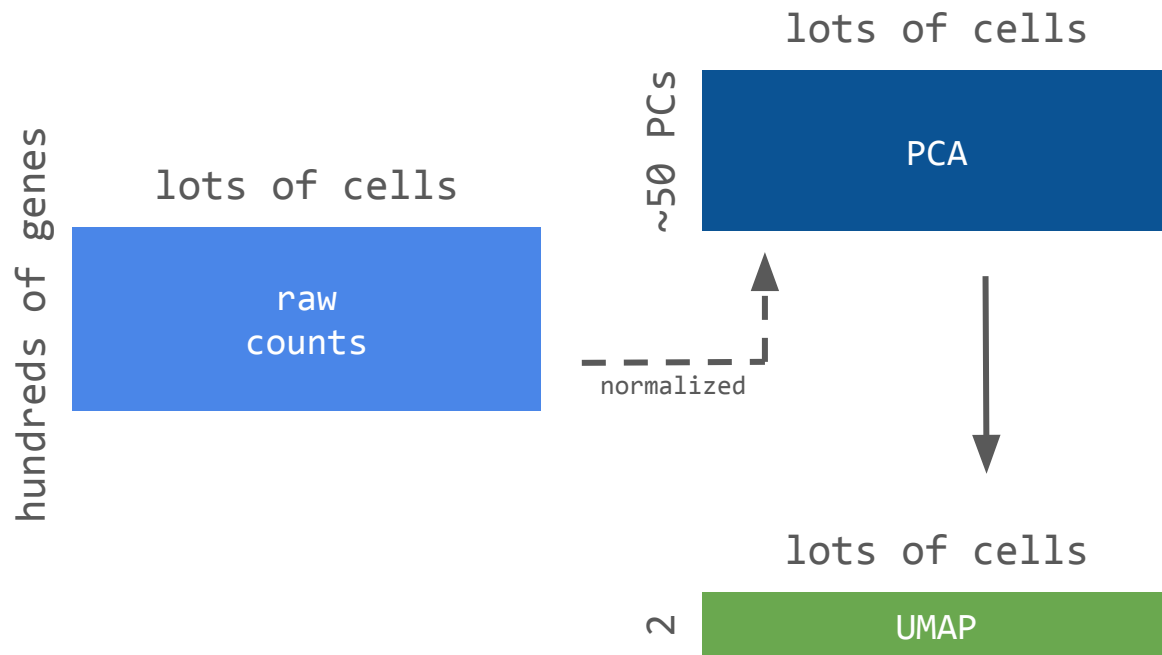
Spatially Informed Clustering

Spatially Informed Differential Expression

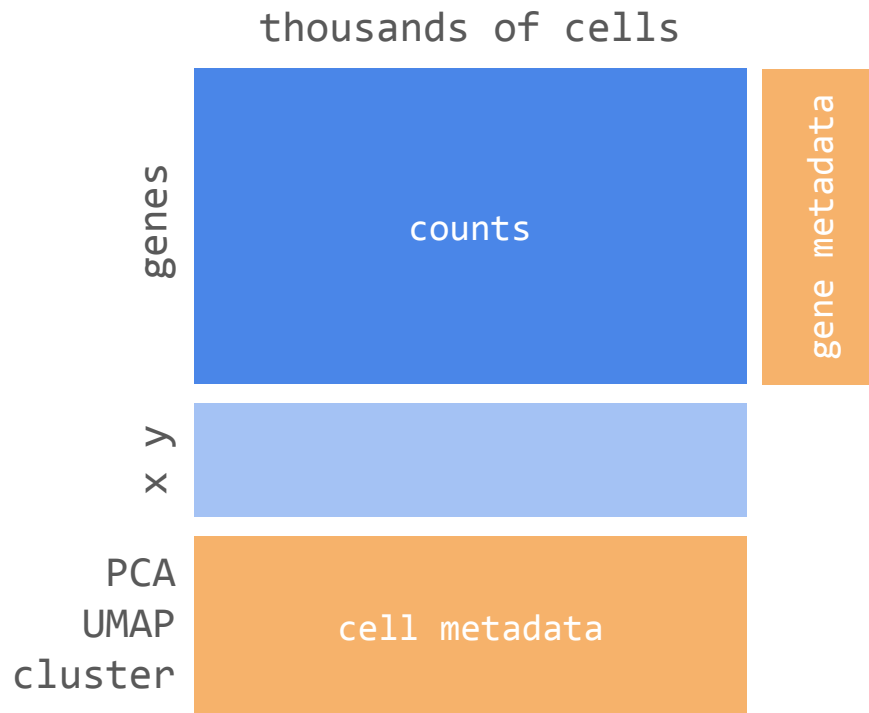
Dimensionality Reduction



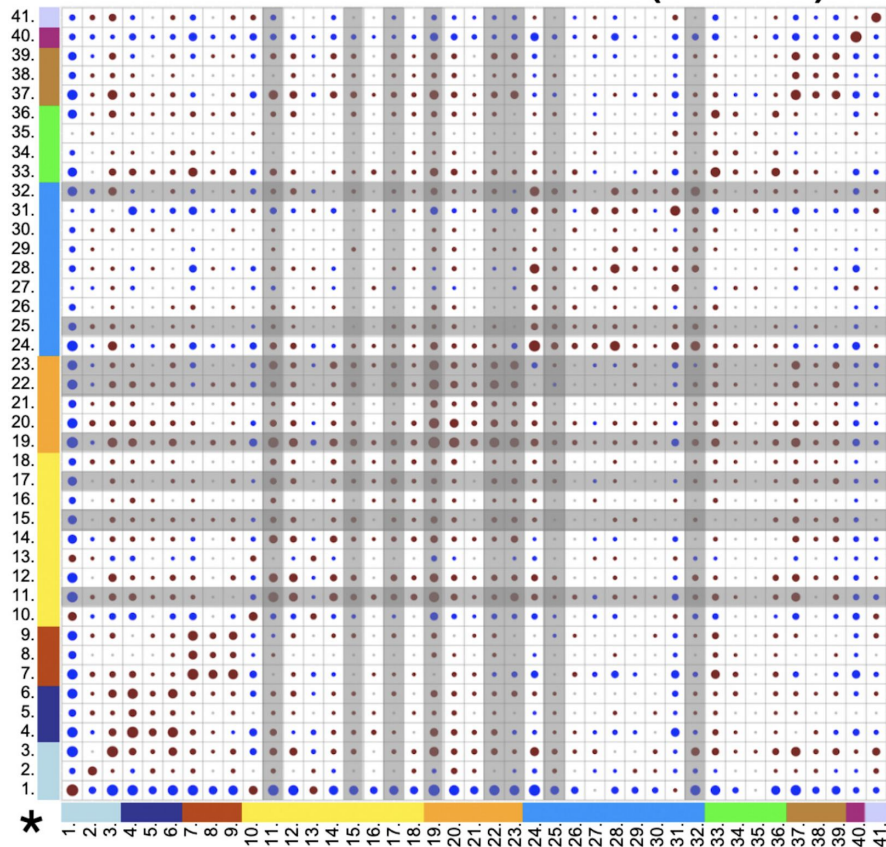
Dimensionality Reduction



Spatial Transcriptomics Analysis



Attraction/Avoidance



```
{imcRtools}
```

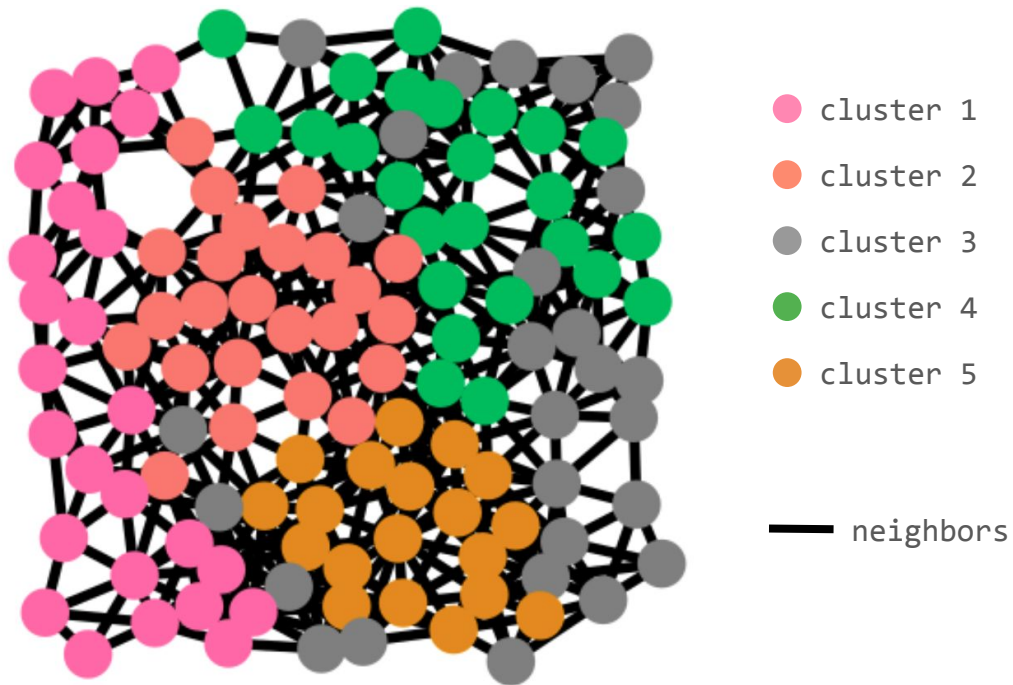
```
testInteractions()
```

For each pair of cell types, do they tend to co-localize or not?



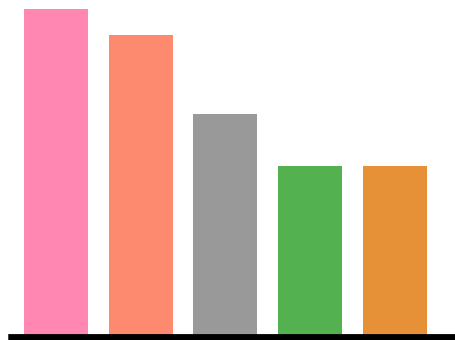
Neighborhood Analysis

Spatial Layout



Spatial Genomics

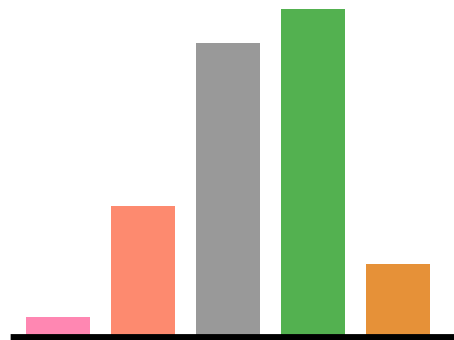
All Cells



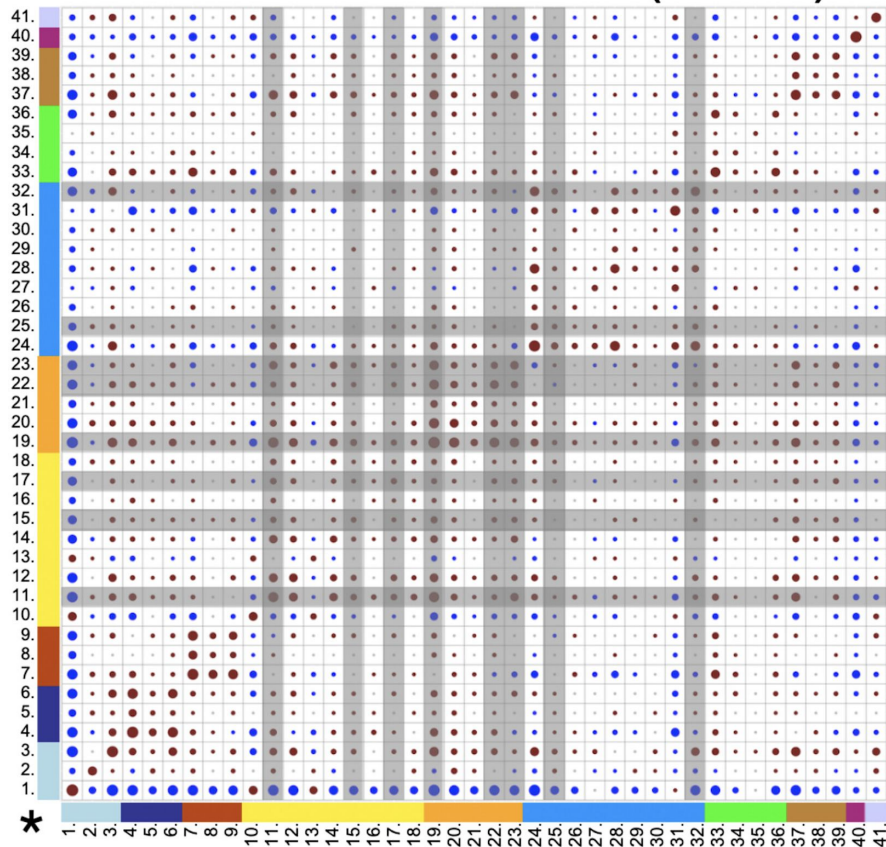
Null
distribution

Neighbors of

● cluster 4



Attraction/Avoidance



```
{imcRtools}
```

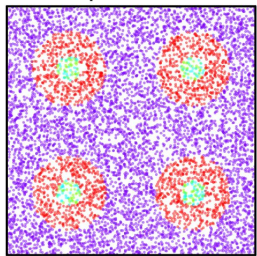
```
testInteractions()
```

For each pair of cell types, do they tend to co-localize or not?

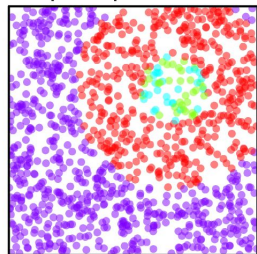


Multi-Scale Co-Localization

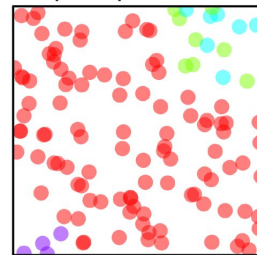
Full spatial extent



750 μ m spatial extent



250 μ m spatial extent



● A

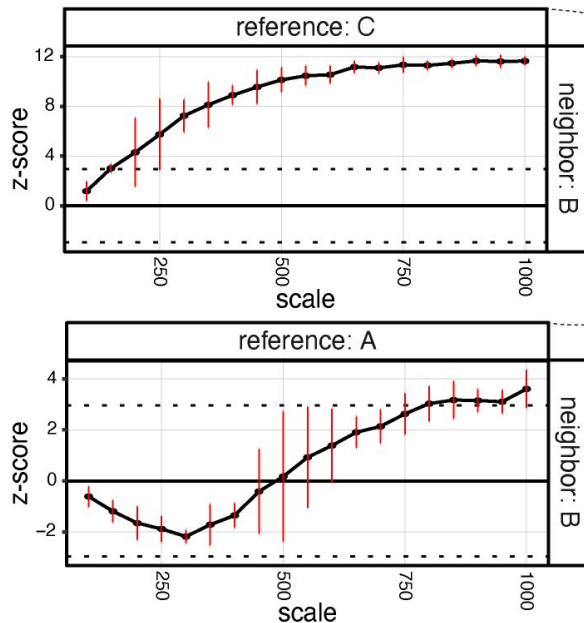
● B

● C

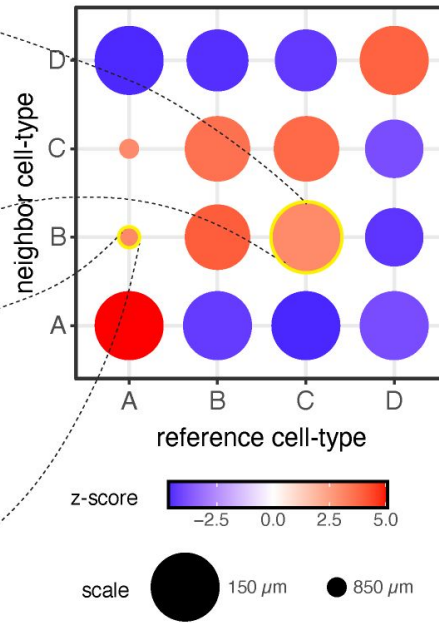
● D

Cell Type

multi-scale cell-type
spatial relationship trends



multi-scale cell-type
spatial relationship summary



Spatial Integration

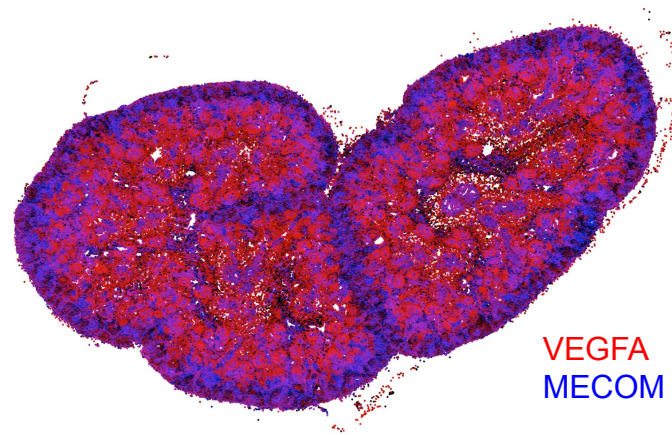
Spatial Integration

scRNAseq



+

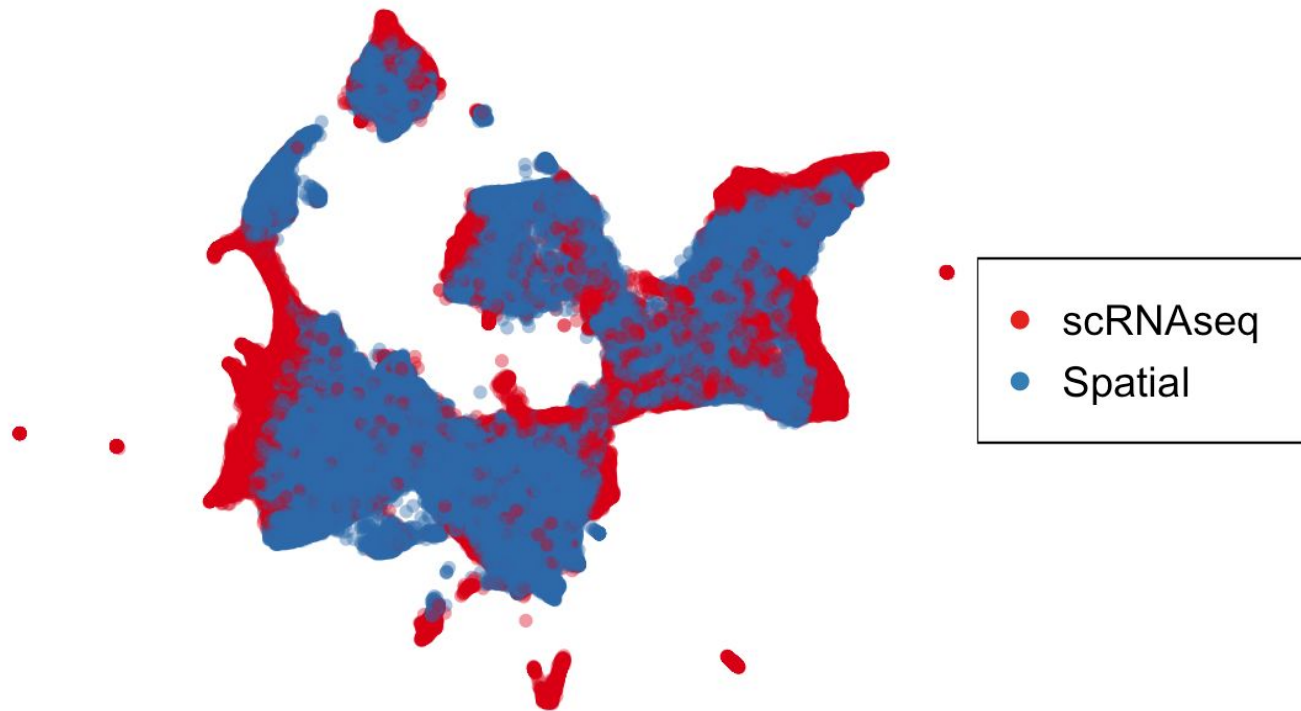
Spatial
Transcriptomics



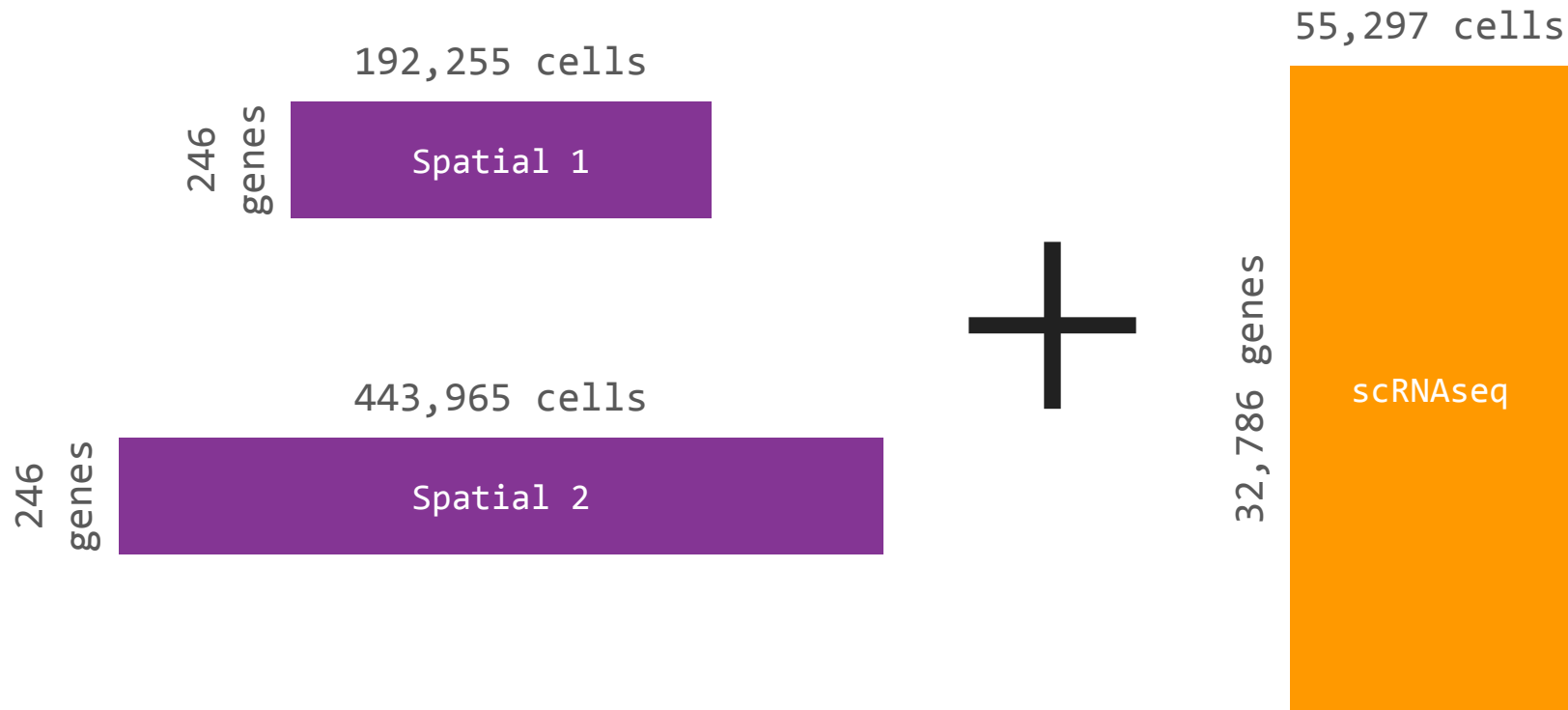
VEGFA
MECOM

Spatial Integration

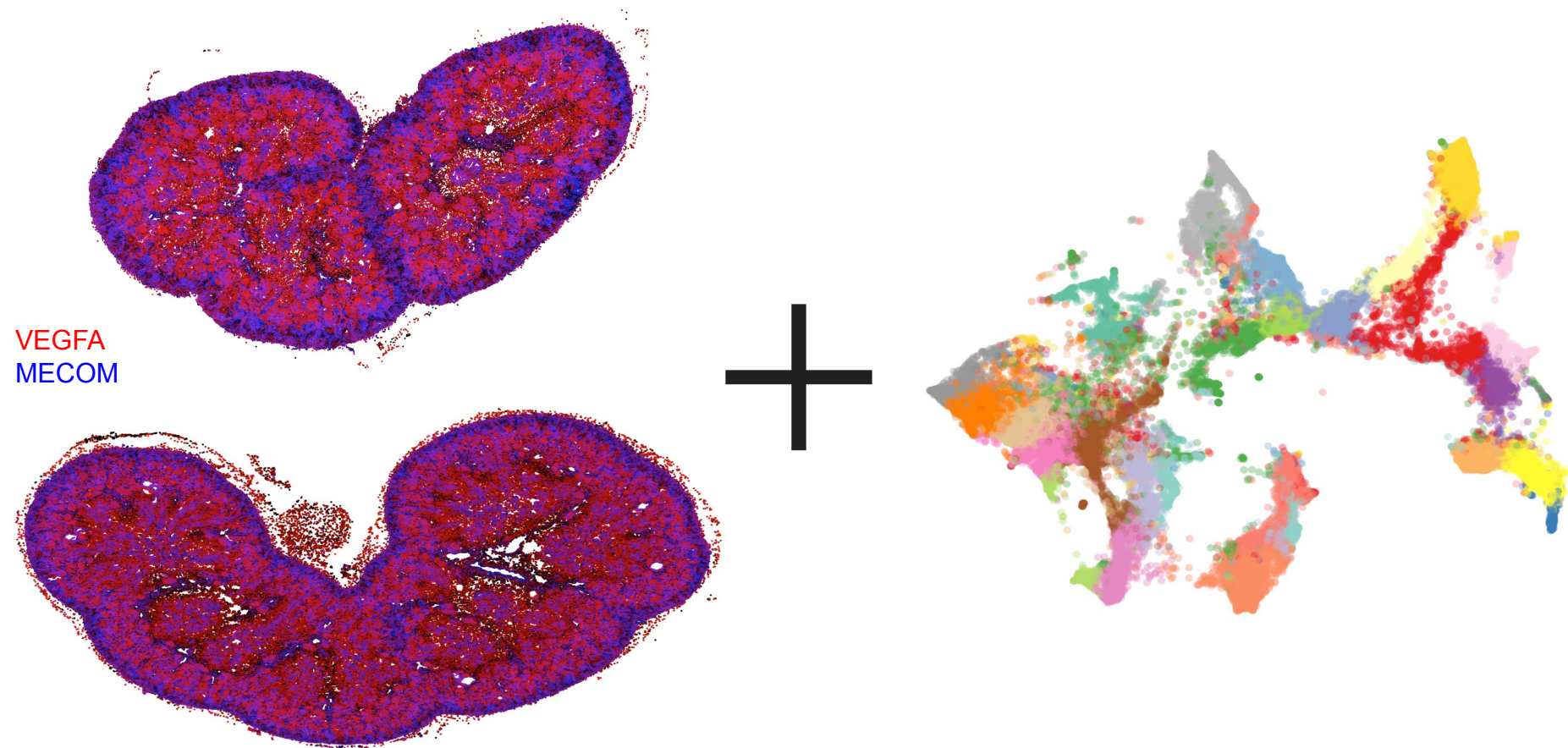
Combined UMAP



Spatial Integration

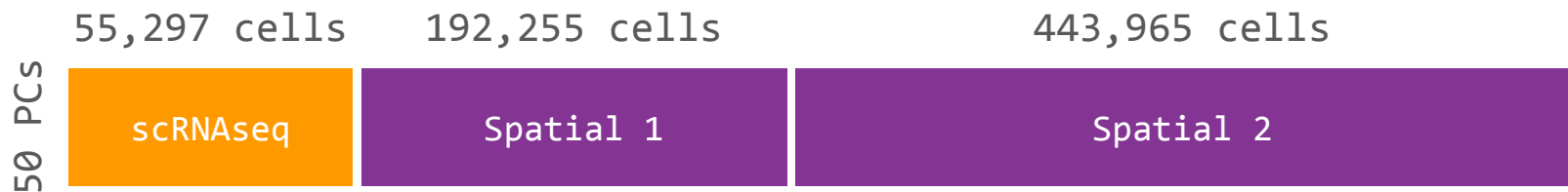


Spatial Integration



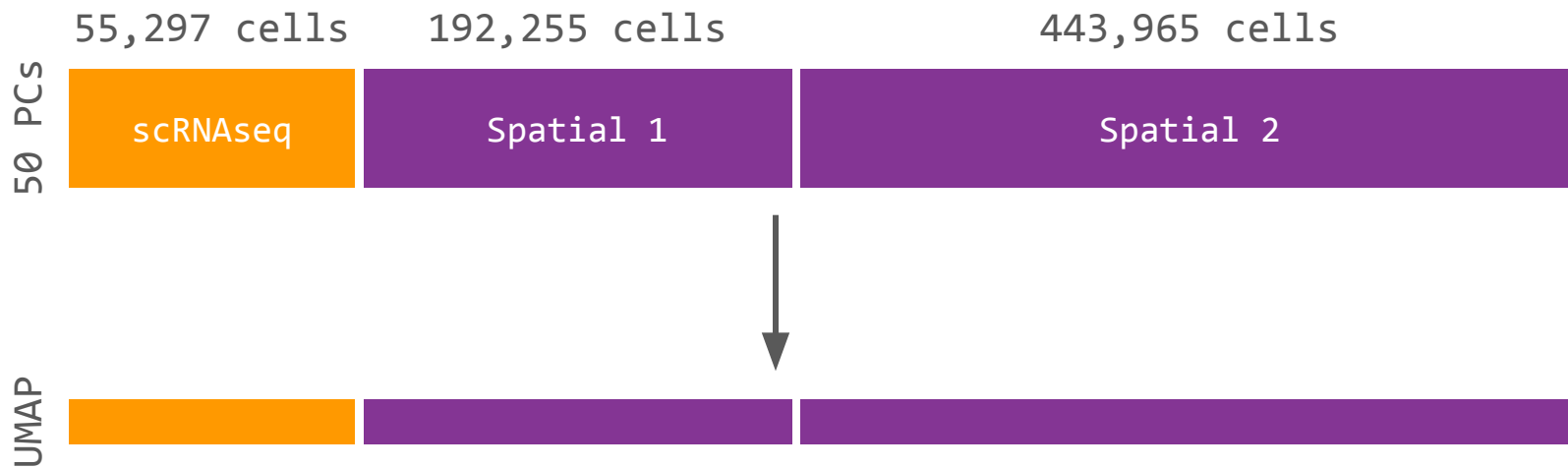
Spatial Integration

{StabMap}



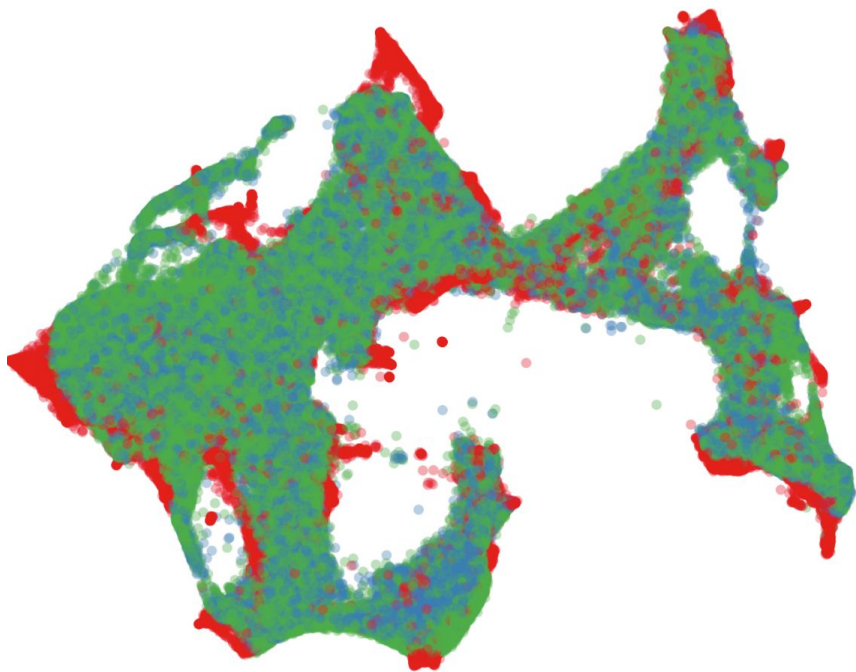
Spatial Integration

{StabMap}



Spatial Integration

Combined UMAP



scRNA-seq



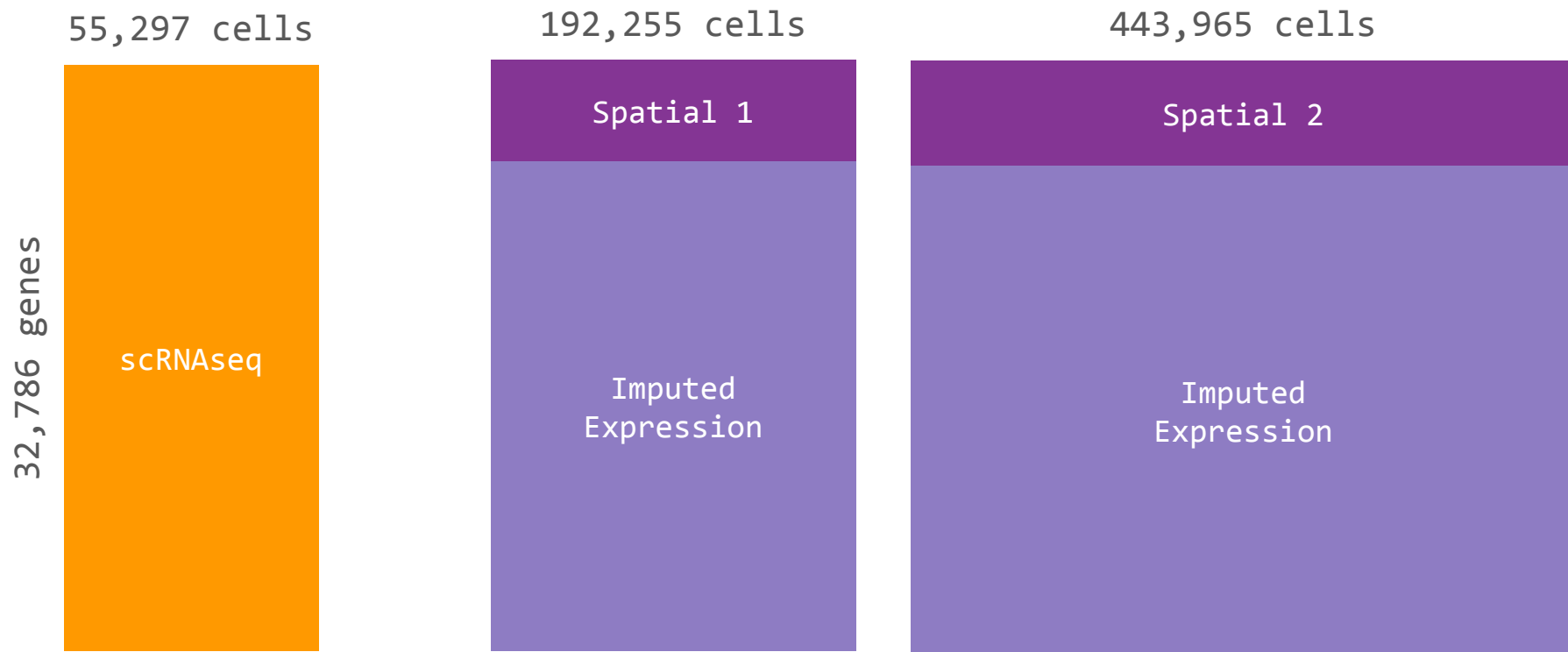
Spatial 1



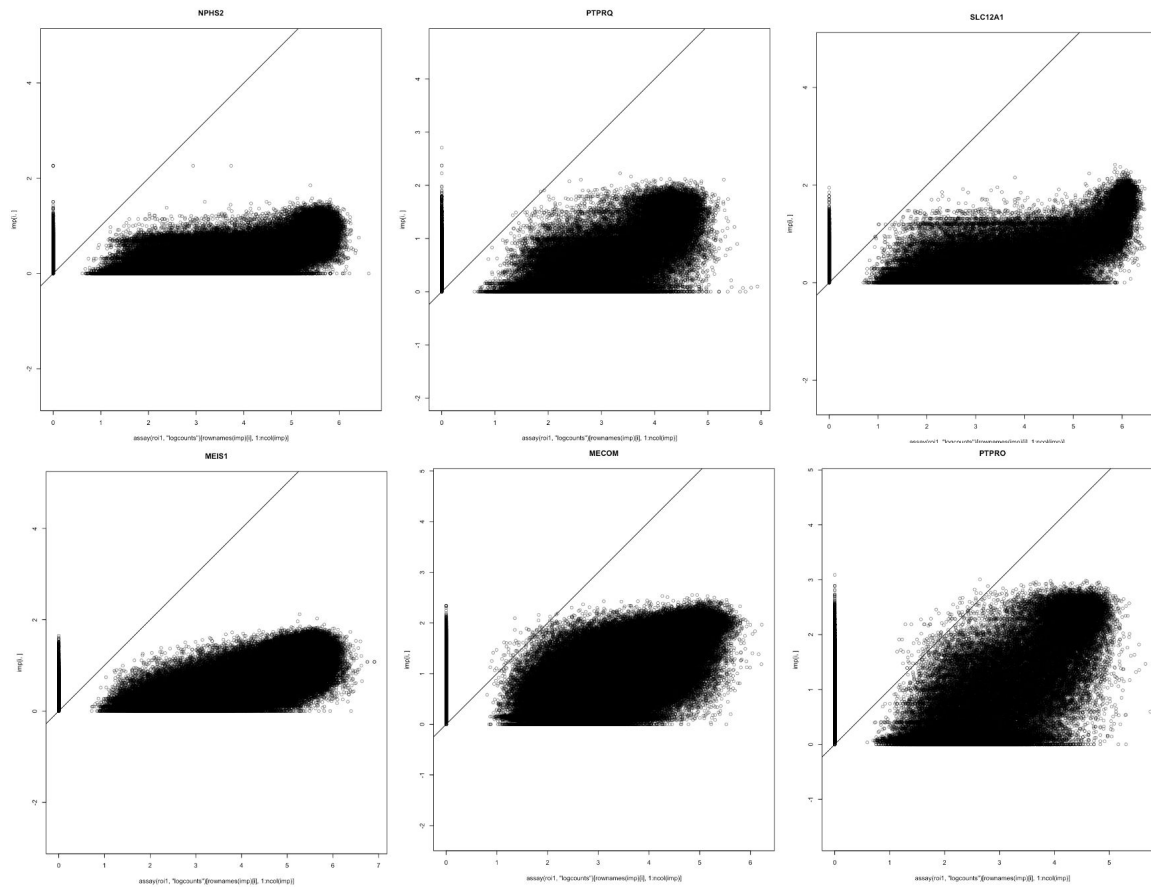
Spatial 2



Expression Imputation

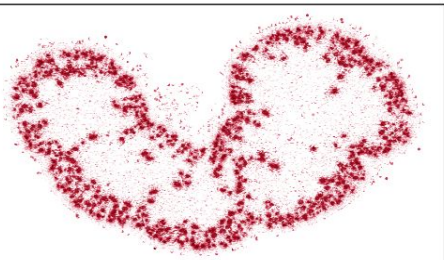


Expression Imputation

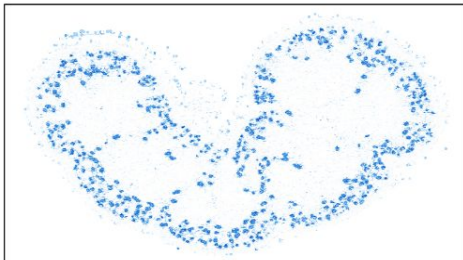


Expression Pattern Imputation

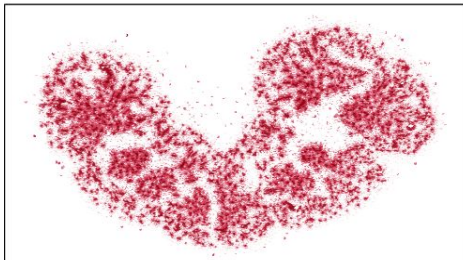
NPHS2 - Actual



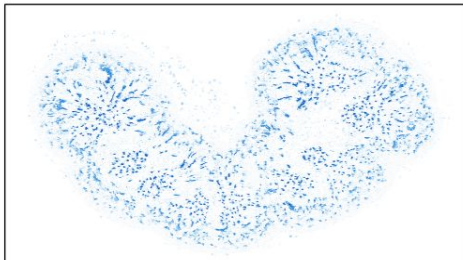
NPHS2 - Imputed



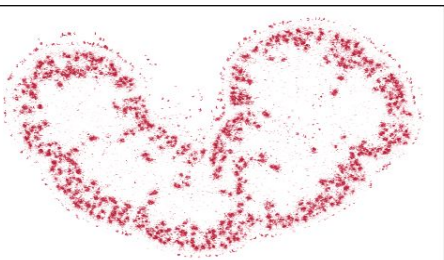
SLC12A1 - Actual



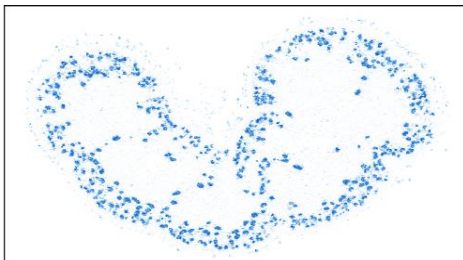
SLC12A1 - Imputed



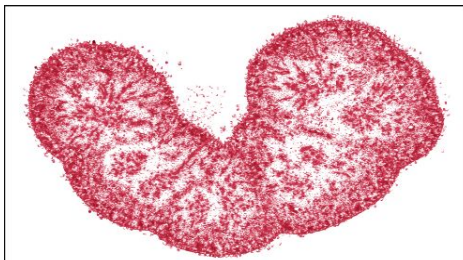
PTPRQ - Actual



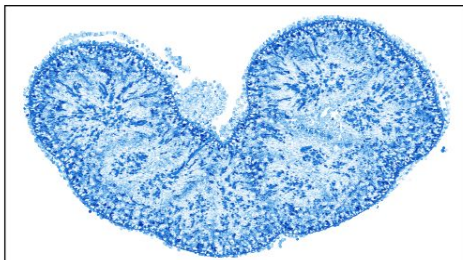
PTPRQ - Imputed



MECOM - Actual



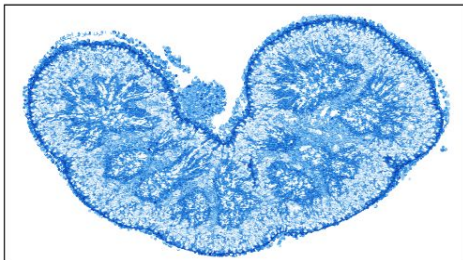
MECOM - Imputed



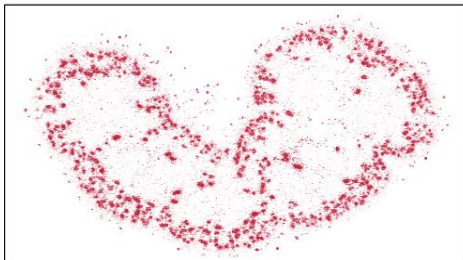
MEIS1 - Actual



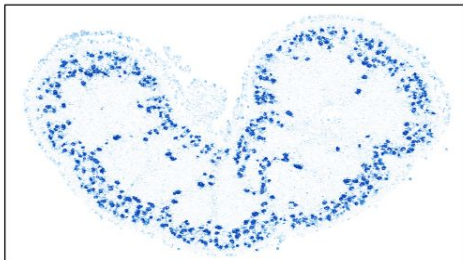
MEIS1 - Imputed



PTPRO - Actual

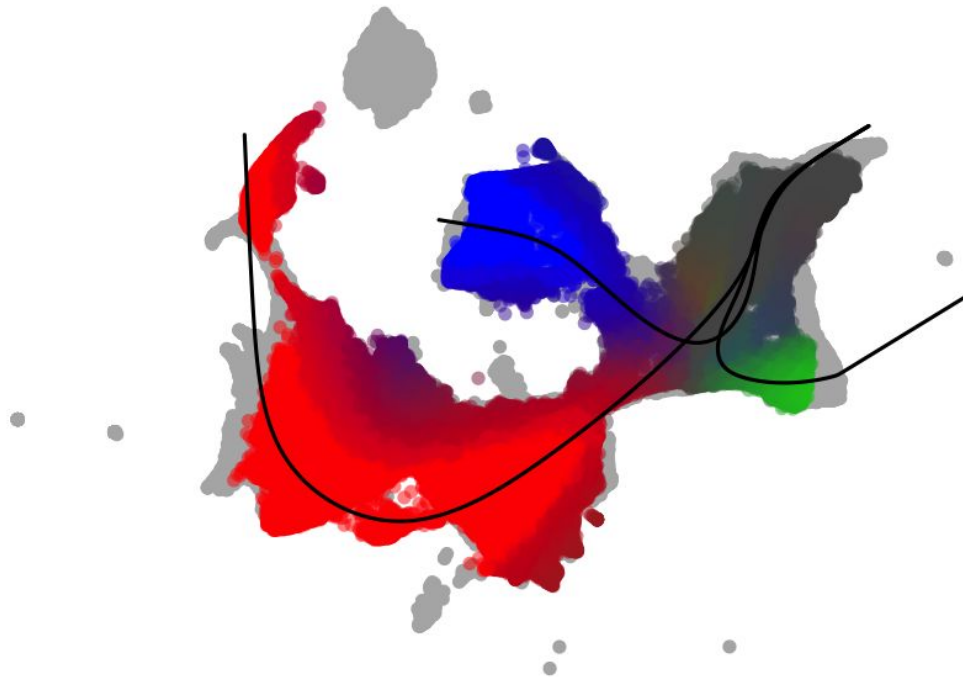


PTPRO - Imputed



Pseudotime Mapping

Combined UMAP



Pseudotime Mapping

